

Constructing the Source Code of the Universe: From the Tachyon Sea to a Theory of Everything

Bin Xin[#]

Shaoxing SeaLens Technology Co., Ltd., Shaoxing 312000, China

[#]Corresponding author: xinbin280@163.com

<https://orcid.org/0000-0002-5884-5124>

Abstract: Contemporary fundamental physics is mired in a profound paradigmatic crisis. Its root lies in an irreconcilable fissure within the underlying logic of existing theoretical paradigms: the mainstream approach incoherently juxtaposes the "speed-of-light limit" and "quantum entanglement non-locality" as two disconnected, mutually un-derivable primitive empirical facts. To bridge the resulting chasm between theory and observation, physics has increasingly devolved into a patchwork of "ad hoc physical imagery"—concepts such as the forced segregation of matter and antimatter in Plasma Cosmology, the recursive nesting of universes in Black Hole Cosmology, the perpetual fission of realities inherent in the Many-Worlds Interpretation, and the fine-tuned survivability posited by the Anthropic Principle are continuously introduced and stacked, often contrived to satisfy specific mathematical forms rather than physical necessity. This mode of development resembles performing inconsistent functional updates atop an ever-growing "spaghetti code" base; while capable of temporarily fitting data through fine-tuning, it has led physics into a dead end by utterly sacrificing the simplicity, inevitability, and direct descriptive power that theories ought to possess, driving the field toward infinite involution and away from physical reality. We assert that any patchwork within the logic of the old paradigm will only lead us further astray from the true physical picture. Therefore, this paper advocates for and implements a radical paradigm reconstruction. Starting from the foundational principles upon which physics rests—such as **reality**, **energy-information conservation**, **evolutionary unitarity**, and **non-singularity**—we aim to write a new, logically closed "Source Code of the Universe." Through a critical review of the existing theoretical "source code," we identify and rectify a key implicit constraint: that the establishment of quantum entanglement must be subluminal. By rigorously relaxing this constraint, we derive for the first time a logical necessity for the universe's ground state—a superluminal, globally entangled "Tachyon Sea." This ground state is governed by a **PT-symmetric non-Hermitian Hamiltonian** that guarantees unitary evolution, operating on a dynamically adaptive spacetime substrate generated by its own dynamics. Within this framework, the entire phenomenological structure of our observable universe—its spacetime, causality, speed limit, and matter properties—naturally and inevitably emerges as a local, decohered "Classical Bubble." The fundamental goal of this paper is not to provide a phenomenological model, but to rigorously demonstrate that an intrinsically consistent, self-referential "Universal Algorithm" can naturally "emerge" from a unified logical origin, fundamentally reconciling all isolated contradictions of the old paradigm. Specifically, this framework offers a novel, internally consistent physical picture for resolving the foundational conflict between quantum mechanics and general relativity. It provides a common interpretive basis for competing quantum gravity paths represented by String Theory (internal vibrational modes) and Loop Quantum Gravity (external oscillatory geometry). Under this new philosophy of physics, the common origin of spacetime and matter (driven by the zero-energy condition), the nature of dark matter (high-dimensional tachyon projections) and dark energy (boundary geometric effects), the resolution of the black hole information paradox (via tachyonic phase transition returning to the Tachyon Sea), cyclic cosmic evolution, and even the geometric origins of particle identity, spin, and fundamental forces (with symmetries like $SU(3)$ and $SU(2)$ interpreted as projections of higher-dimensional spherical vibration modes), are no longer ad hoc puzzles requiring external introduction. Instead, they are the inevitable manifestations of a single set of "Universal Source Code" operating self-consistently across different logical strata.

1 An Interpretation of the Emergence of Space and Time

Recent advances in holographic duality — the Ryu-Takayanagi formula, the entanglement wedge construction — show that quantum entanglement in a boundary field theory can fully encode the geometry of higher-dimensional gravitational spacetime [1, 2]. The constraint that the entanglement wedge must always contain the causal wedge further suggests that entanglement is not merely a passive participant but the fundamental source of spatial geometry [3]. To understand our universe—an expanding spacetime with macroscopic twistability and microscopic discreteness—it is necessary to combine entanglement with decoherence processes [4].

1.1 Emergence of Space: From Entanglement Network to Classical Geometry

Space is not a pre-existing continuous manifold but an effective geometry emergent from a Planck-scale quantum entanglement network.

Entanglement defines proximity: In holographic duality (AdS/CFT), the mutual information $I(A:B) = S_A + S_B - S_{AB}$ between two subregions A and B in the boundary quantum field theory is proportional to the minimal surface area of the intersection of their corresponding entanglement wedges [1]. This implies that the spatial “distance” between two points is determined by the strength of quantum entanglement between them. For the global pure state $|\Psi\rangle_{\text{univ}}$, the entanglement strength $E_{ij} = S(\rho_i) + S(\rho_j) - S(\rho_{ij})$ between any two Planckian units i, j is monotonically anti-correlated with the effective geometric distance $d(i,j) \sim f(1/E_{ij})$ [5]. Strong entanglement corresponds to proximity; weak or zero entanglement corresponds to remoteness or even disconnected components.

Quantum network model: At the Planck scale $l_P \sim 10^{-35}\text{m}$, spacetime is hypothesized to be a network composed of qubits (or more general quantum degrees of freedom). Each node represents a Planck volume unit, and edges between nodes represent entanglement correlations. The total entanglement structure of the entire network uniquely defines the intrinsic geometry of the universe. Continuity and smoothness of space are emergent results of the average over large-scale entanglement.

Unification of twistability and discreteness: The twisting of spacetime described by the classical Einstein field equations (already confirmed by gravitational wave experiments), as well as the inevitable quantum nature at the Planck scale (e.g., the discrete area spectrum in loop quantum gravity) and “viscosity” (the pinning effect of energy-momentum changes on the spacetime structure), can all be regarded as macroscopic manifestations of the collective response of the underlying quantum entanglement network. [6] The inhomogeneous distribution of the entanglement network directly gives rise to curvature; the response delay and dissipation of the network manifest as viscosity.

Gravitational waves: The Einstein equations can be derived from the first-order variation of entanglement: for a holographic boundary subregion A, the variation of entanglement entropy δS_A with respect to shape deformation is related to the energy-momentum tensor via the Bianchi identity [5]. This means that macroscopic curvature corresponds to inhomogeneous distribution patterns in the entanglement network; gravitational waves are the propagation of collective excitation modes of the network, which, upon quantization, become gravitons.

Cosmic expansion = increase in the number of spatial quanta: In standard cosmology, expansion is described by the growth of the scale factor $a(t)$. However, in the discrete quantum network picture, expansion should be reinterpreted as an increase in the number of quanta: the total volume of the universe is $V(t) = N(t) \cdot V_0$, where V_0 is the size of a single Planck volume (the fundamental quantum) and remains constant; $N(t)$ grows with time. Expansion does not stretch the

fundamental units but adds network nodes and entanglement edges. The total entanglement entropy $S_{\text{total}}(t) \propto N(t)$ increases with expansion and always satisfies the holographic entropy bound $S \leq A/(4l_p^2)$ [2].

Classical space is stabilised by an “entanglement crust” and continuous decoherence. Decoherence locks in a definite macroscopic background. An isolated quantum network is in a superposition of multiple geometries; after coupling with the environment, decoherence suppresses the coherence between different geometries, causing collapse to a classical spacetime. Holographic duality shows that introducing noise (simulating decoherence) into a boundary CFT subregion reduces the recoverable region on the gravity side from the entanglement wedge to the causal wedge [7], indicating that decoherence indeed causes the loss of some geometric information, and what remains is the classically accessible spacetime.

1.2 Emergence of Time: Information Recording/Updating within a Reference Frame

Unitary evolution and microscopic reversibility: In an isolated quantum system, the dynamics of quantum entanglement are described by the unitary transformation $U(t)=e^{-iHt}$. This process preserves entropy and is in principle reversible. Therefore, time-reversal symmetry holds at the microscopic level, with no distinction between past and future. This is the starting point for the emergence of time — the arrow of time is not an inherent property of fundamental laws.

Decoherence generates event records: When the network couples to the environment (e.g., all high-energy degrees of freedom not explicitly monitored), environment-induced decoherence diagonalizes the system’s density matrix and suppresses quantum interference between different geometric configurations. Each time the system interacts with the environment, a copy of the system’s state is left in the environment, forming an indelible “event record.” The accumulation order of these records defines the direction of time.

Non-unitary mapping and the arrow of time: Classical time (the irreversible, directed arrow of time) does not come from microscopic unitary evolution; however, unitary evolution provides the basis for microscopic reversibility, while the irreversibility of time originates from information recording during decoherence. Formally, the decoherence process of a subsystem is described by a completely positive and trace-preserving map $\varepsilon(\rho_s) = \sum_k E_k \rho_s E_k^\dagger$, where the Kraus operators E_k are non-unitary. The discrete time step is written as $\rho_s^{t+1} = \varepsilon_t[\rho_s^t]$. Since tracing out the environment is irreversible, the composition order of these maps is non-commutative, thereby giving rise to the thermodynamic arrow of time.

Time as the order of information records: Therefore, the essence of time is the unidirectional accumulation of environmental memory. Without decoherence, there is no classical time; without an environment, there is no definite past or future. The irreversibility of **classical time = accumulation/updating of information records**: In the classical universe, every macroscopic event (e.g., stellar collisions, particle detections) leaves behind a large number of environmental memories (photons, thermal radiation, etc.). The unidirectional increase of these memories forms an irreversible causal order, which is the arrow of time we experience. Time is not a fundamental background stage, but rather a “running log” generated by the continuous “updating” of quantum information within the cosmic system, readable by internal observers.[8] The universe is a dynamic quantum information network. The “present” is the consensus, unalterable log record within the network; “time flow” is the continuous updating of the system’s state due to irreversible interactions (decoherence) between quantum information and the environment.

1.3 Phenomenological Interpretations

In this framework, the universe is viewed as an open quantum system, its “global log” consisting of a sequence of irreversible update events; the subjective time of each observer corresponds to the frequency and sequence with which they synchronize with the global log. The following reinterprets five key phenomena using this framework.

Twin Paradox: The “Relative Log” of Macroscopic Time and Its Resynchronization

The younger brother on Earth stays synchronised with the dense “cosmic master log”. The elder brother travels away at high speed; the spaceship becomes a quasi-closed “local information bubble”. Due to the Doppler shift and relativistic spacetime structure, the rate of information exchange between the spaceship and the Earth environment is greatly reduced (i.e., time dilation). During the cruising phase, the elder brother’s personal log evolves independently, updates slowly, and becomes desynchronized from the master log. The key is the turnaround and return: the spaceship must forcibly break its original inertial connection and undergo a violent information resynchronization with the Earth environment. This instantaneous “log merger” allows the elder brother to receive, in an extremely short time (subjectively very brief), the long history of updates accumulated by Earth. Consequently, the elder brother’s total log entries are far fewer than the younger brother’s, and upon reunion, the younger brother is older. Conclusion: the paradox arises from differences in the pace of information updates between inertial frames. The irreversible synchronization during the turnaround breaks the symmetry, providing an intuitive explanation without invoking acceleration details.

The “Timelessness” of Photons and Light-Speed Travelers: Zero Update Means Zero Proper Time

Special relativity predicts that a particle moving at the speed of light (e.g., a photon) has zero proper time. This means that from the photon’s perspective, the creation and annihilation of the universe occur simultaneously. In the information-update interpretation, this counterintuitive conclusion receives a direct and natural explanation. The source of updates is interaction: the passage of time corresponds to continuous information exchange between the system and its environment. For a freely propagating photon in vacuum, its evolution is described by the unitary operator $U(t)=e^{-iHt}$, which exchanges no information with the environment; the photon’s polarization, momentum, and other degrees of freedom remain unchanged, and no event occurs that can be externally recorded. Since no log entry is written, from the photon’s “subjective perspective” the number of time steps it experiences is zero. By extension, any entity moving at the speed of light — whether an elementary particle or a hypothetical “light-speed spaceship” — must satisfy the condition of zero information exchange with the environment. The moment one attempts to exchange information with it (e.g., send a signal), the entity no longer remains at light speed. Hence, “timelessness” is a universal feature of all light-speed movers, rooted physically in the absence of information updates rather than merely kinematic relations.

Wheeler’s Delayed-Choice Experiment: Postponed Log Submission and Post-Measurement Causal Fixation

Wheeler’s delayed-choice experiment reveals a puzzling feature of quantum mechanics: an observer can decide whether to measure “particle-like” or “wave-like” behavior after a photon has already passed through the double slits, and the measurement outcome seems to retroactively change the photon’s past path. The Copenhagen interpretation abandons a realist description of “the past,” while the information-update interpretation offers an alternative clear language. Unitary stage: computation without recording. From emission to arrival at the detection screen, the photon’s

evolution is governed by the Schrödinger equation's unitary dynamics. During this stage, the photon state is a path superposition, where the phase information includes all experimental apparatus settings (including later inserted measurement devices). This evolution is fully reversible: if all measurement devices are removed before the photon arrives, the system can cleanly return to its initial state. Importantly, no information has been irreversibly written into the environment — it is like a computer program performing multi-path calculations in the background without committing the final result. Measurement as log submission. When the photon reaches the detection screen and is recorded (e.g., generating an observable current via the photoelectric effect), an irreversible information update occurs. The environment (detector atoms, electronic circuits, etc.) acquires definite eigenstate information. At that moment, the system must select one of the many computational paths as “factual history” and formally write it into the cosmic log. The key to the delayed-choice experiment is that the type of measurement apparatus (whether a which-path detector is placed, whether a beam splitter is inserted, etc.) determines which observable's eigenvalue (position or momentum) is “submitted.” Because the unitary stage already contains all possible experimental configuration information, the final log entry is entirely self-consistent with the measurement setup, producing no logical contradiction. The illusion of “retrocausality.” Observers feel that “the past has been changed” because they are accustomed to thinking that history is fixed before measurement. In the information-update interpretation, before measurement (log submission), there is no definite classical history — only the correlation potentials described by unitary dynamics. Measurement does not discover a pre-existing fact but creates a new log entry. This entry is compatible with the subsequent classical causal chain but is independent of any “pre-measurement reality.” Thus, no need to invoke action-at-a-distance or backwards causation; it is sufficient to recognize that the moment of information update is the moment history solidifies. The delayed-choice experiment merely illustrates a nontrivial instance of this principle[8].

Creation and Annihilation of Virtual Particles: “Potentiality” vs. “Actualization” of Microscopic Time Updates

From an information perspective, the creation and annihilation of a virtual particle-antiparticle pair is a fully reversible quantum computation: the system starts from the vacuum, evolves into particle states, and then evolves back to the vacuum, ending in the same state. Because no energy or information leaks into the environment, the entire process leaves no permanent record in the cosmic log — it is merely a “virtual trial run” of the vacuum, contributing no passage of time. If, before annihilation, the virtual particles are perturbed by the environment (e.g., a black hole horizon, a strong electric field, or scattering with a detector), the particle information irreversibly diffuses into the environment and solidifies into real particles. This event creates a new entry in the global log, marking an irreversible solidification of a fact. Therefore, each “virtual-to-real” event constitutes a fundamental unit of microscopic time update in the universe. The statistical accumulation of many such events recovers the thermodynamic arrow of time, whereas fully annihilated virtual pairs are like cancelled computations and do not propel time forward.

Quantum Entanglement and “Faster-than-Light” Correlations: Holistic Collaborative Update, Not Signal Transmission

An entangled state is an indivisible quantum information whole. Measuring particle A triggers a global information update — because this update occurs within one time, the entire system's state is atomically reconstructed. The state of particle B is not “notified”; it is automatically determined by the global update. This instantaneity is not signal transmission but the inherent cooperativity of

the update: instantaneity = update cooperativity. No information is transmitted across spacelike intervals, so relativity is not violated.[5]

2 Constructing an Alternate World—The Tachyon Sea

2.1 A Thought Experiment: The Fast Universe

The fundamental asymmetry is not decoherence itself but: **The speed of quantum disentanglement is faster than light ($\gg c$), The speed of establishing quantum entanglement is limited by the speed of light ($\leq c$).** This asymmetry breaks time-reversal symmetry at a more fundamental level than decoherence, providing a physical basis for irreversibility that is more primitive than decoherence. Decoherence is merely the macroscopic emergent manifestation of this asymmetry when an open system couples to its environment. In the universe's quantum information processing, "disconnecting" is faster than "connecting." This asymmetry drives the unidirectional accumulation of log entries, thereby generating the passage of time that we perceive.

Analogous to the symmetric relationship between entanglement and disentanglement in our world, we define the Doctor Xin's rule for this fast universe:

The speed of establishing quantum entanglement is faster than light ($\gg c$);

The speed of quantum disentanglement/decoherence is limited by the speed of light ($\leq c$).

Since time arises from decoherence while our universe follows unitary evolution, causality is preserved despite superluminal interactions (no time reversal occurs). To comply with this rule, the elementary particles in this universe must have imaginary mass ($m^2 < 0$), and these particles always travel faster than light[9]. Based on the above rules, we speculate on the properties of spacetime and matter in the fast universe: Matter consists of tachyons with imaginary mass, with energy-momentum relation $E^2 = p^2 c^2 - |m^2| c^4$, and speed always exceeding the speed of light.

Properties of space: Space is not a pre given smooth background but emerges from an entanglement network. The "distance" between two tachyons is determined by the strength of entanglement (stronger entanglement means shorter distance), rather than by coordinate differences. Space is essentially a discrete, arbitrarily connectable graph structure. Properties of time: Any signal travels far faster than light. There is no globally consistent temporal order. In the fast universe, there is no absolute "past" or "future"; only decoherence events provide local "before/after" relations that propagate at the speed of light. As time emerges from the decoherence process. This world appears as an infinitely entanglement: countless tachyons flash into existence instantaneously, then form a dynamic network through global entanglement. Entanglement in the network is slowly broken by local decoherence wavefronts ($\leq c$); after being broken, the tachyons become randomly entangled again, forming new connections. The overall picture is: fast generation of correlations (instantaneous) + slow erasure of correlations (at light speed), leading to incessant "bubble like" fluctuations. There are no fixed particle trajectories, no eternal structures, only transient entanglement clusters that constantly appear and disappear. The core of the fast universe is: a globally coherent boiling soup + locally decoherent bubbles[10]. "Emergence" is driven by the difference in timescales between entanglement and decoherence. Entanglement establishment is instantaneous (timescale ≈ 0), while decoherence propagation requires at least $\Delta t \geq \Delta x / c$. Thus, a cluster of tachyons first instantly "flashes" into an entangled structure, then after some time (depending on the size of the region), the decoherence wavefront sweeps through and breaks it apart. This is like bubbles continuously rising in a soup: rapidly generated, slowly dissipated. "Clustering" is

not about spatial proximity, but about entanglement proximity. In our world, “clustering” means being close in spatial position. But in the fast universe, even if two tachyons are billions of light years apart, as long as their degree of entanglement is sufficiently high, they are considered “in the same bubble.” Thus, clustering is clustering on the entanglement network, not spatial co location. Decoherence only operates within a “light speed accessible” range. This rule is crucial: for two tachyons to undergo decoherence (i.e., to break their mutual entanglement), they must satisfy $\Delta x \leq c \cdot \Delta t$ spatially. In other words, decoherence is a local, slow erasure mechanism. Consequently: Long distance entanglements are never disturbed by decoherence $\rightarrow$ they form long range stable correlations that constitute the “background structure” of the fast universe. Short distance entanglements are constantly broken by decoherence and then reestablished $\rightarrow$ they produce local dynamic fluctuations, i.e., those “boiling bubbles.” There is no global time in the entire world, only local temporal order provided by local decoherence events. Because there is no light cone causal restriction, the whole universe cannot be described by a single time coordinate. Each decoherence event (propagating at the speed of light) defines a “before/after” within its range of influence, but there is no absolute order between different events.

... Wait a moment... this locally encapsulated decoherence region – is that not exactly our classical universe? Our ordinary world (the slow universe) is precisely a “decoherence bubble” within the fast universe.

When a decoherence event occurs in some region of the fast universe, the event propagates a decoherence wavefront outward at the speed of light. Inside the wavefront, global instantaneous entanglement is destroyed, and all interactions and signal propagation are limited to the speed of light c , thereby giving rise to the familiar physical rules of the slow universe (speed-of-light limit, causal order, locality, real-mass particles, etc.). This region inside the wavefront is a “decoherence bubble,” which is exactly our slow universe (our own universe). The bubble wall is the decoherence wavefront. Inside the bubble is an ordered, light-speed-limited local world; outside the bubble is the chaotic, instantaneously entangled boiling soup of the fast universe. Hence, the slow universe and the decoherence bubble in the fast universe are one and the same entity. We live inside the bubble and can only perceive physical laws below the speed of light. This counter-intuitive ground state — a globally entangled condensate of imaginary-mass fields — is what we call the *tachyon sea*.

2.2 Tachyon Sea Hypothesis of Cosmic Evolution

2.2.1. The Matrix: A Sea of Tachyons with Imaginary Mass

The ground state of the universe is a timeless, boundaryless quantum gravitational system whose fundamental degrees of freedom consist of a tachyon field with imaginary mass squared ($m^2 < 0$). [9, 10] This tachyon sea resides in a false vacuum state of extremely high energy density, obeys unitary evolution as a whole, and possesses global quantum entanglement. In this matrix, there exist no classical time or space dimensions; all physical degrees of freedom are in coherent superposition states.

2.2.2. Decoherence and the Birth of Our Universe

Due to quantum fluctuations, a spontaneous decoherence event occurs in some local region of the tachyon sea. Decoherence causes this region to condense from the false vacuum to the true vacuum, releasing vacuum energy. This process is accompanied by a topological phase transition:

real-mass modes ($m^2 > 0$) emerge from degrees of freedom that originally only had imaginary mass, forming “tachyon-geometry” bound states (Planck-scale bubbles). The bubble wall expands outward at the speed of light. Inside the bubble, the decoherence process defines a local arrow of time (the radial direction of the bubble wall) and a three-dimensional space (the region enclosed by the bubble wall). Simultaneously, a speed-of-light limit and causality are established inside the bubble. A classical observable universe (a “slow universe”) forms inside the bubble.

2.2.3. Cosmic Expansion and Evolution

The continuous expansion of the bubble wall is equivalent to cosmic inflation and the subsequent accelerated expansion of space. The matter inside the bubble (real-mass particles, radiation) evolves according to the rules described by general relativity and quantum field theory: gravity dominates the formation of large-scale structures, while electromagnetic, strong, and weak forces govern microscopic processes. As expansion proceeds, the recession speeds between different matter regions may exceed the speed of light, but as long as they lie inside the same bubble, they remain causally connected. Cosmic evolution follows the second law of thermodynamics, irreversibly evolving from a low-entropy initial state to a high-entropy state.

2.2.4. Causal Isolation Leads to Particles Leaving Our Universe and

Dissolving into the Tachyon Sea

When the universe expands to a sufficiently large scale, the recession speed of some matter regions exceeds the speed of light, causing them to move beyond each other’s causal horizons. In the Big Freeze scenario, the horizon eventually engulfs everything, and every particle ends up at the center of its own horizon, forever losing contact with the outside. At that point, the quantum entanglement between these mosaic regions and the main body of the universe is severed, and they become causally isolated systems. Their superluminal behavior precisely corresponds to the superluminal motion in the tachyon sea. **Phase transition trigger:** For a causally isolated system containing only real-mass particles, its wavefunction undergoes a fundamental change.[11] Within the framework of quantum gravity, such a system can evolve into a spacelike mode via tunneling or analytic continuation, i.e., its effective mass squared changes from positive to negative ($m^2 > 0 \rightarrow m^2 < 0$). The newly generated tachyons possess the same quantum properties as the parent tachyon sea and naturally become re-coherent. The information carried by the original particles is encoded into the global entanglement structure of the tachyon sea, achieving non-local storage of information. The entire evolution is unitary, information is conserved, and energy is preserved.

2.2.5. Cycle and Generation of New Bubbles

After receiving information and energy from causally isolated particles, the local fluctuation conditions in the tachyon sea are replenished. New decoherence events randomly and continuously occur in the matrix, forming new Planck bubbles. Each new bubble independently undergoes the complete process of birth, expansion, causal isolation, and return to the matrix. Different bubbles are causally separated and invisible to each other. Our observable universe is just one of many transient “slow-universe bubbles.” The bubble wall expands outward at the speed of light, so the tachyon sea outside the bubble can never be detected by observers inside the bubble.

3 Tachyon Dynamics Equations

In relativistic quantum mechanics, a tachyon with imaginary mass ($m^2 < 0$) is a relativistic object whose evolution is described by the Klein-Gordon (KG) equation:

$$\frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} - \nabla^2 \phi = \frac{m^2 c^2}{\hbar^2} \phi$$

When $m=i\mu$ is imaginary, the right-hand side becomes $-\frac{\mu^2 c^2}{\hbar^2} \phi$. This is itself the standard form of the tachyon field equation. The general solution of the imaginary-mass KG equation usually diverges exponentially ($e^{\pm \kappa t}$), corresponding to unstable tachyon condensation. To obtain a stable equation that is unitary, we need to impose closed-space boundary conditions. In a closed space (e.g., a box), boundary conditions quantize the wavenumber, allowing oscillatory solutions.

3.1 Dynamics Equations in Different Dimensions

3.1.1 One-dimensional case

Consider a one-dimensional tachyon of length L in the interval $[0, L]$ with Dirichlet boundary conditions $\phi(0,t) = \phi(L,t) = 0$.

$$\frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} - \frac{\partial^2 \phi}{\partial x^2} = -\kappa^2 \phi, \quad \text{with } \kappa = \frac{\mu c}{\hbar} > 0$$

Separation of variables: let $\phi(x,t) = X(x) T(t)$.

Spatial part: $X_n(x) = \sin(k_n x)$, with $k_n = \frac{n\pi}{L}$, $n=1,2,3,\dots$

Time part: $\frac{1}{c^2} \ddot{T}_n = (k_n^2 - \kappa^2) T_n$.

Criterion: If $k_n > \kappa$, the right-hand side is positive, $\ddot{T}_n = +\omega_n^2 T_n$, solution $T_n \sim e^{\pm i\omega_n t}$ oscillatory in time, stable and unitary. If $k_n < \kappa$, the right-hand side is negative, solution $T_n \sim e^{\pm \kappa_n t}$ exponential divergence (unstable). To keep tachyons stable, we cut off the low-frequency (long-wavelength) modes, ensuring that only modes with $k_n > \kappa$ exist in the system. This implies:

Physical realisation: the spatial length L must be smaller than some critical value.

$$L < L_{\max} = \frac{\pi \hbar}{\mu c}$$

Alternatively, for a fixed length L , the imaginary mass μ must be sufficiently small to satisfy $\mu < \pi \hbar / (Lc)$, so that even the lowest mode $n=1$ satisfies the oscillation condition. In a one-dimensional closed space, the long-wavelength (low-frequency) instability of the imaginary-mass tachyon is cut off by the boundary conditions, leaving only short-wavelength, high-frequency stable oscillatory modes. This physical picture is analogous to a vibrating string in string theory under an imaginary-mass background.

3.1.2 Three-dimensional spherical symmetry

Dynamics in three dimensions and the Planck-scale issue. Stability criterion in the three-dimensional spherically symmetric case: consider an imaginary-mass field ϕ inside a sphere of radius R with homogeneous Dirichlet boundary condition $\phi(r=R,t)=0$. Write $\phi(\mathbf{r},t)=\psi(r)T(t)$. The radial part uses spherical Bessel functions. To ensure the time part yields oscillatory behaviour: the

radial equation for the spherically symmetric case (with angular momentum quantum number $\ell=0$ – the lowest mode, also the most dangerous because it is non-singular at $r=0$ and has the longest wavelength):

Radial equation: $\frac{d^2 u}{dr^2} + (k^2 + \kappa^2)u = 0$, Where $u(r) = r R(r)$, $R(r)$ radial wave function k is the separation constant (real wavenumber). The equation becomes

$$\frac{d^2 u}{dr^2} + K^2 u = 0 , K^2 = k^2 + \kappa^2$$

The solution is $u(r) = A \sin(Kr)$ (finite at $r=0$).

Boundary condition $u(R)=0$ gives $KR=n\pi$, $n=1, 2, 3$. i.e.

$$k^2 + \kappa^2 = \left(\frac{n\pi}{R}\right)^2$$

The time part $\ddot{T} = -c^2 k^2 T$. The key is that k^2 must be positive for oscillatory solutions:

$$T(t) \sim e^{\pm i c |k| t}, k^2 = \left(\frac{n\pi}{R}\right)^2 - \kappa^2, \text{ which means } \left(\frac{n\pi}{R}\right)^2 > \kappa^2.$$

The most dangerous is the lowest mode $n=1$, so stability requires $\pi/R > \kappa$. Thus the critical radius is

$$R < R_{max} = \pi/\kappa = \pi \hbar / (\mu c). \text{ The Planck mass is } m_P = \sqrt{\frac{\hbar c}{G}} \approx 2.18 \times 10^{-8} \text{ kg} , \text{ Planck length } l_P =$$

$$\sqrt{\frac{\hbar c}{c^3}} \approx 1.62 \times 10^{-35} \text{ m}. \text{ Set } \mu = \alpha m_P \text{ where } \alpha \text{ is dimensionless. Then } R_{max} = (\pi \times l_P) / \alpha:$$

If $\alpha = \pi$ (i.e., $\mu \sim \pi m_P$), then $R_{max} = l_P$ is Planck scale.

If $\alpha > \pi$ (imaginary mass larger than Planck mass), $R_{max} < l_P$, which is meaningless in current physics.

If $\alpha < \pi$ (imaginary mass smaller than Planck mass), $R_{max} > l_P$ and can be much larger.

Therefore, the critical radius is not necessarily of the order of the Planck scale; it depends on μ .

If μ is small, macroscopically stable “imaginary-mass tachyons” can exist in large spaces.

3.1.3 General d-dimensional evolution equation and stability criterion

Considering that a sphere has the largest geometric “size”, in a d -dimensional space ($d \geq 3$) with a spherical cavity of radius R , the imaginary-mass field equation is

$$\frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} - \nabla_D^2 \phi + \kappa^2 \phi = 0 , \phi(r=R) = 0 \text{ with } \kappa = \frac{\mu c}{\hbar}$$

The stability criterion (unitary evolution condition) – that no mode diverges, i.e., all modes are oscillatory in time – is that the eigenvalue of the lowest angular momentum radial mode is greater than κ^2 . The condition is

$$\left(\frac{z_d}{R}\right)^2 > \kappa^2$$

where z_d , is the first zero of the d -dimensional radial Bessel function for zero angular momentum.

Critical radius formula:

$$R_{max}(d) = \frac{z_d}{\kappa} = \frac{z_d \hbar}{\mu c}$$

Variation of critical radius with dimension

z_d is the first zero of the d -dimensional spherically symmetric Bessel function (zero angular momentum). Its numerical value increases monotonically with dimension d . [11, 12]

Dimension d	Radial equation type	First zero z_d (approx.)	Critical radius R_{crit}
---------------	----------------------	----------------------------	----------------------------

3	$j_0(z)=0$	$\pi \approx 3.1416$	$\pi \hbar/(\mu c)$
4	$J_{1/2}(z)=0$	≈ 3.8317	$\approx 3.83 \hbar/(\mu c)$
5	$j_1^{(5)}(z)=0$	≈ 4.493	$\approx 4.49 \hbar/(\mu c)$
6	$J_{2/3}(z)=0$	≈ 5.1356	$\approx 5.14 \hbar/(\mu c)$
d (even)	$J_{d-2/2}(z)=0$	$\sim \frac{d}{2} + O(d^{-1/3})$	increases with d
d (odd)	spherical Bessel zero	$\sim \frac{d}{2} + a\left(\frac{d}{2}\right)^{1/3} + b + O(d^{-1/3})$	increases with d

Conclusion: $R_{max}(d)$ increases monotonically with d . Higher dimensions can accommodate larger stable imaginary-mass fields.

High-dimensional limit and physical implications

No upper bound on dimension: From a purely mathematical perspective, d can be any positive integer. For any d , as long as the radius $R < R_{max}(d)$, the imaginary-mass field inside the d -dimensional sphere is stable and oscillatory, and the evolution is unitary.

Asymptotic behaviour for large d : When $d \rightarrow \infty$, the first zero z_d has the asymptotic form

$$z_d \sim \frac{d}{2} + \frac{1}{2^{2/3}} \alpha_1 \left(\frac{d}{2}\right)^{1/3} + \dots$$

where $\alpha \approx 2.338$ is the first zero of the Airy function. This means that in very high dimensions, the critical radius grows linearly with d : $R_{max} \sim \frac{d}{2} * \frac{\hbar}{\mu c} O\left(d^{\frac{1}{3}}\right)$. Extremely high dimensions: the stable region can be very large. The “hyperspherical Bessel functions” oscillate faster in higher dimensions, so the first zero occurs at a larger argument, allowing a larger cavity radius without triggering the exponential instability (radial energy is converted into angular energy).

If the imaginary mass is $\mu = \alpha m_P$, then

$$R_{max}(d) = \frac{z_d}{\alpha} l_P$$

For fixed μ , R_{max} increases with d . Higher dimensions make it easier to avoid being compressed below the Planck scale. Only when $z_d \lesssim \alpha$, would $R_{max} \lesssim l_P$ approach the Planck scale. For large d , this would require μ to be very large (a large α).

Conclusion: No upper bound on dimension: d can be any positive integer, mathematically fully self-consistent. Higher dimensions are more “tolerant”: $R_{max}(d)$ increases with d . Higher dimensions suppress the long-wavelength instability of the imaginary-mass field more strongly, allowing larger closed spaces. Unified evolution: in all dimensions, the field is expanded as a superposition of hyperspherical Bessel functions $j_l^{(d)}(kr)$ and time phases $e^{\pm i\omega t}$, as long as $R < R_{max}(d)$, the evolution is unitary. Counter-intuitive discovery: for an imaginary-mass field, increasing the spatial dimension actually enlarges the scale of stable existence, rather than confining it to the Planck scale.

3.2 Perturbations

The spatial evolution of tachyons under small perturbations, and the shape selection effect: the stability of the system is determined by the “minimum characteristic scale”, i.e., oscillation along the “narrowest” direction. Consider a three-dimensional rectangular box with dimensions L_x, L_y, L_z with $L_x \ll L_y, L_z$. Stability criterion: the wavenumber must satisfy $k > \kappa$. The smallest wavenumber is determined by the smallest dimension $k_{min} \approx \pi/L_x$ (because L_x is smallest). The Dominant mode is

need to satisfy $k > \kappa$, the system must rely on the narrowest direction x . Therefore, the lowest-frequency vibration mode (the fundamental mode) indeed appears as a standing wave along the narrowest direction, while being approximately constant (zero mode or very low-order mode) along the other directions y & z . With the physical picture: the instability (exponential divergence) of the imaginary-mass field corresponds to long-wavelength (low-frequency) perturbations. The narrowest dimension naturally cuts off the long wavelengths, like a pair of scissors trimming away the unstable “long waves”, forcing the system to oscillate at shorter wavelengths (higher frequencies). Tachyon vibration forms: in the lowest-order approximation they appear as “strings”, but in the full picture they are “branes”.

One dimension (string): In one dimension (e.g., a line segment), the vibration is simply the standard string vibration: $\phi(x,t) \sim \sin(n\pi x) e^{\pm i\omega t}$. This is the “open string” picture.

Two dimensions (membrane): The situation becomes more complex. In a two-dimensional rectangular region, if $L_x \ll L_y$, the lowest mode is indeed $\phi(x,t) \sim A \sin(\pi x / L_x)$. This looks like a “string” vibrating in the x direction. However, once higher modes are excited, mixed standing waves in both x and y directions appear (e.g., $\sin(\pi x / L_x) \sin(\pi y / L_y)$), which are membrane vibrations.

Three dimensions and higher (supermembrane): No longer simply a string. In a sphere or a rectangular box of dimensions $d \geq 3$: Fundamental mode (lowest mode): due to symmetry or the minimum-size constraint, it may appear as a one-dimensional vibration in one direction (e.g., the spherically symmetric $\ell=0$ mode, which depends only on r). Excited states: once the angular momentum $\ell > 0$, the vibration simultaneously extends over multiple dimensions, manifesting as complex multi-dimensional standing waves, described by hyperspherical Bessel functions and hyperspherical harmonics.

Considering the d -dimensional sphere space of mathematical essence, Separated variables:

$$\phi(\mathbf{r},t) = R(r)Y(\Omega)e^{\pm i\omega t}$$

Time part $T(t) \sim e^{\pm i\omega t}$ always oscillatory – this is the “string-like” temporal behaviour. Radial part $R(r)$: satisfies an equation of the form $d^2\mu/dr^2 + K^2\mu = 0$. This equation is mathematically identical in form to the one-dimensional string equation after a suitable transformation. Angular part $Y(\Omega)$: this is the “extra decoration” brought by higher dimensions. The “string vibration” essence is a mathematical manifestation of the radial equation. The field’s oscillation in the radial direction (from centre to boundary) obeys the same laws as a one-dimensional string, but the whole system is also overlaid with complex angular structures.

In the d -dimensional spherical cavity model, tachyon stability is achieved through a balance of external confinement and internal support. Externally, the Dirichlet boundary condition acts as a quantum trap, discretizing wave numbers and cutting off long-wavelength unstable modes that would otherwise cause exponential divergence. Internally, the centrifugal barrier generated by angular momentum counters the inherent inward collapse driven by the negative mass. Stability is only realized when these geometric contributions collectively overcome the tachyon's imaginary mass, transforming its behavior from explosive growth to bounded oscillation and ensuring unitary evolution within the finite volume. Considering the lowest energy levels and perturbative effects, the lowest-energy spatial configuration for stable tachyons should be a d -dimensional ellipsoid. The instability is determined by the smallest characteristic length (the radius $r_{\min}$ along the shortest axis). Along that shortest axis, the radial part behaves as a one-dimensional string vibration, and its energy exactly cancels the intrinsic tachyon energy. The angular part takes the form of multi-dimensional standing waves on a $(d-1)$ -dimensional membrane. In the case of an isolated

space with no heat exchange, the interior “string” undergoes an eternal “pulsation” (contraction and expansion) in the radial direction. The surface “membrane” (with $\ell > 0$) undergoes eternal “pattern oscillations” in the angular directions.

3.3 Cutoff: Dimensional Matching and Imaginary Mass Matching

The spontaneous locking of the system’s dimension emerges from the competition between two opposing tendencies: a) Geometric energy minimization – lower-dimensional configurations generally require less “bending energy” or tension to maintain, meaning lower dimensions correspond to lower intrinsic geometric energy costs. b) Dynamical stability maximization – higher-dimensional spaces support a richer spectrum of angular vibrational modes (higher-dimensional spherical harmonics), which more effectively suppress the intrinsic instability of the tachyon field (the exponential divergence driven by imaginary mass). In other words, higher dimensions enhance the system’s robustness against perturbations and fluctuations, leading to a smaller variance in its energy state. Therefore, the optimal dimension results from a trade-off between “reducing energy cost” and “enhancing stability.” In what follows, we use this conceptual framework to present a series of physically motivated, though mathematically qualitative, arguments regarding the system’s energy composition, mass cut-off, and symmetry constraints.

3.3.1 Imaginary-mass cutoff:

a) Lower bound – although mathematically μ can be arbitrarily small, physically this is meaningless. Planck cutoff ($\mu \gtrsim M_{Pl}$): Effective field theory breaks down when $\mu \ll M_{Pl}$. The characteristic scale of the system, $R \sim 1/\mu$, becomes much larger than the Planck length. In this case, the “flexible space” effectively degenerates into a large, flat classical background, quantum gravity effects (the flexibility of space itself) become negligible, and the tachyon instability is trivialised, losing the “self-organising” geometric effect. Physical meaning: in this model, μ must be of the order of M_{Pl} or larger, so that space remains “compact and flexible” and the geometric tension can compete strongly with the tachyon instability. Otherwise, this physical image will become a classical universe.

b) Upper bound – When the imaginary mass becomes too large, the system tends to “split” into multiple smaller stable units, rather than being compressed indefinitely. In essence, the instability of energy is resolved by a topological change of the spatial structure (the mechanism of tachyon condensation and spatial fragmentation). Limit of a single cavity: when κ (the absolute value of the imaginary mass) is extremely large, the radius R required for stability becomes extremely small ($R < \pi/\kappa$). However, R cannot be arbitrarily small – it is limited by the Planck scale or by diverging energy density. The total surface area of two small spheres is larger than that of a single large sphere, providing more “surface vibration modes” to dissipate or store energy, thereby reducing the energy density. “Very large imaginary mass” corresponds to a deep tachyon potential in physics. Single-point instability: at one point, the potential $V(\phi)$ is negative with a large absolute value, and the field strongly tends to roll toward some non-zero vacuum expectation value $\langle \phi \rangle = \pm v$. Domain wall formation: space spontaneously forms domain walls, with $\phi = +v$ on one side and $\phi = -v$ on the other. Spatial partitioning: these domain walls physically act as new spatial boundaries, dividing the original space into multiple regions.

3.3.2 Dimensional matching:

To match the tachyon energy, space needs to possess a “self-organising” flexibility. The tachyon model eventually evolves into: a) **Dimension emergence**: starting from chaotic initial conditions, the system spontaneously locks onto a specific dimension d_{opt} by adjusting (the parameters d and R). b) **Stable vibration**: in this dimension, the imaginary mass of the tachyon is exactly balanced by the centrifugal barrier and geometric tension, and the system enters a PT-symmetric, rhythmically rotating vibrational state.

The system (spatial energy defined positive here) minimises the total energy by adjusting d to balance the “tachyon instability” against the “cost of the high-dimensional centrifugal barrier”. The effective potential V_{eff} roughly consists of two competing parts:

$$V_{eff}(d, R) \approx -\kappa^2(d) + \frac{C(d, l)}{R^2} + \frac{D}{R^{d-1}} V$$

Tachyon term (negative energy): often scales with d (e.g., in string theory, $\kappa^2 \sim (d_{crit})$). Geometry term (positive energy): includes the centrifugal barrier and spatial bending energy. Higher dimensions have a strong centrifugal barrier, but “driving” so many dimensions itself requires energy (e.g., the tension needed to maintain a high-dimensional volume). If d is too small, the tachyon term κ^2 dominates (large negative), and the system becomes extremely unstable (exponential blow-up). If d is too large, the geometry term (e.g., $C(d, l)/R^2$, where $C(d, l) = l(l + d - 2)$), explodes and the energy becomes too high. There exists an optimal d_{opt} (e.g., 4, 8, ...) such that the total energy V_{eff} takes a global minimum and stable. In this dimension, the system neither explodes nor pays an excessively high geometric maintenance cost.

3.3.3 Joint cutoff of imaginary mass and optimal dimension:

Numerical search d_{opt} : Because the geometric term E_{geom} grows extremely fast with d (e.g., factorial-like from the volume), the total energy typically attains a minimum at a small integer d . If the growth is mild (e.g., linear), the optimum d_{opt} is often between 6 and 12. If the growth is strong (e.g., exponential, which is more reasonable), the optimum d_{opt} falls between 4 and 6. This model is a classic “energy minimisation” problem, and the result is usually low single-digit dimensions like $d=4,5,6$.

Reasonable parameter range (natural units):

Parameter	Determining mechanism	Reasonable range
Tachyon imaginary mass μ	Planck cutoff (to avoid effective field theory breakdown)	$1 \lesssim \mu \lesssim 10$ (order of Planck mass M_{Pl})
Emergent space dimension d	Energy minimisation (competition between geometric energy and tachyon energy)	4, 5, 6
Spatial radius R	Stability boundary (Planck-length order)	$R \sim D/2\mu \sim 1$

The logical chain as: μ is large $\rightarrow R$ is small $\rightarrow$ high-dimensional geometric energy increases sharply $\rightarrow$ suppresses unbounded growth of $d \rightarrow$ the optimal d_{opt} is finite and small.

3.4 System Symmetry

The symmetry requirements of the “ d -dimensional spherical cavity + imaginary-mass field + closed system” determine the “vibration patterns” of the field and the rules of evolution.

3.4.1. Core symmetries

The model possesses a hierarchy of symmetries from “global” to “local”:

Symmetry	Mathematical object	Physical meaning	Manifestation in the spherical cavity
Time translation	$t \rightarrow t + \Delta t$	Energy conservation (Hamiltonian H)	Frequencies ω are fixed; overall phase rotation of modes
Global U(1)	$\phi \rightarrow e^{-i\theta} \phi$	“Charge” conservation (if ϕ is complex scalar)	If the initial field is real, this symmetry is absent
Spatial rotation SO(d)	$r, \theta \rightarrow r, \theta'$	Angular momentum conservation	Most important: generates spherical harmonics Y_{lm}
Spatial reflection (P)	$\mathbf{r} \rightarrow -\mathbf{r}$	Parity	Splits L into even (positive parity) and odd (negative parity)
Time reversal (T)	$t \rightarrow -t$	Microscopic reversibility	Sends $\omega \rightarrow -\omega$; combined with P gives PT

3.4.2. SO(d) rotational symmetry (high-dimensional angular momentum)

For three dimensions where $d=3$: symmetry group SO(3), generators L_i ($i=x,y,z$), quantum numbers ℓ, m . For Higher dimensions where $d>3$: symmetry group SO(d), more generators, quantum numbers $(\ell, m_1, m_2, \dots)$. The vibration modes of the field are strictly classified by the total angular momentum ℓ . $\ell=0$ (scalar representation): spherically symmetric; the surface is “dead”; only radial string vibrations. $\ell>0$ (tensor representation): the surface is “alive”; corresponds to vibrations on the $(d-1)$ -dimensional sphere S^{d-1} . Degeneracy: modes with $\ell>0$ are highly degenerate. For example, in 3D, one ℓ corresponds to $(2\ell+1)$ values of m . In higher dimensions, the degeneracy is even larger, meaning many different “surface patterns” share the same vibration frequency. Radial direction is a “string” – an SO(d) singlet: due to SO(d) symmetry, the radial direction r is the only “special direction”. It must be independent of angles. Therefore, the $\ell=0$ fundamental mode is necessarily a purely radial vibration – this corresponds to a “one-dimensional string”. Angular directions are a “membrane” – a non-trivial SO(d) representation: modes with $\ell>0$ have their dynamics in the angular directions. Mathematically, they correspond to eigenfunctions of the Laplace-Beltrami operator on the sphere S^{d-1} . These are no longer “string” vibrations but surface waves on the $(d-1)$ -dimensional hypersphere.

Symmetry forces the decomposition of vibrations into “1d (radial string) $\times$ ($d-1$) (angular membrane)”.

3.4.3 The special role of PT symmetry

In a model with imaginary mass ($m^2<0$) and closed boundaries, PT symmetry is not merely a “nice extra” but a “lifeline” that maintains the physical viability of the system. In the classical universe, the usual case ($m^2>0$): H is Hermitian, eigenvalues $\omega^2>0$, the system is automatically stable. In the tachyon sea ($m^2<0$): H is not Hermitian; eigenvalues ω^2 can become negative (ω imaginary, leading to exponential divergence). Salvation by PT: the closed boundary conditions select discrete wavenumbers k such that when $k > |\kappa|$ (the stability condition $\omega^2 = k^2 - \kappa^2 > 0$), although H is non-Hermitian, PT symmetry is not spontaneously broken, guaranteeing that ω is real and evolution is unitary.

3.4.4 Influence of dimension d

In higher dimensions, the symmetry is correspondingly enhanced:

$d=3$: SO(3) symmetry, angular momentum ℓ is an integer.

$d=4$: SO(4) $\approx$ SU(2) $\times$ SU(2); the symmetry is higher, and mode classification is finer.

$d \rightarrow \infty$: extremely high symmetry; the system approaches mean-field behaviour.

3.4.5 Geometry-matter duality

In the tachyon sea model, because space adapts to the quantum dynamics of tachyons, the vibration modes of the tachyon are identical to the vibration modes of the enveloping space. The “particles” and “space” that observed are not independent entities. Internal perspective (string theory / particle physics): one sees high-energy, imaginary-mass “tachyons” vibrating. External perspective (loop quantum gravity / geometry): one sees the violent vibration of the geometric envelope that contains them. The two are equivalent: they are two descriptions of the same physical reality, just like the “shape” and the “vibration modes” of an object.

A stable tachyon picture is as follows: a small sphere with a diameter close to the Planck scale, an imaginary mass between 1 and 10 Planck masses, and vibrating inside a compact, nearly ellipsoidal space of 4–6 dimensions. Symmetry forces the decomposition of spatial vibrations into “1d (radial string) $\times$ (d-1) (angular membrane)”, described by hyperspherical Bessel functions and hyperspherical harmonics. This structure forms what we term a “T-yolk-shell” system. The high-energy tachyon constitutes the internal “T-yolk,” which is the true source of energy and vibration. This yolk is enclosed by a dynamic “T-shell” — the compactified space boundary. The excited graviton resides on this T-shell. Since the total energy $E_{total}=0$, the graviton and the tachyon are tightly coupled inside space. Externally, only changes in geometry and volume are manifested; these vibrational information are encoded on the spatial surface of T-shell. The graviton observed from outside the space does not emit energy, but only transmits information, and the evolution is unitary.

4. Conversion between Imaginary and Real Mass

4.1 Tachyon condensation from imaginary mass to real mass (from the tachyon sea to our universe)

In string theory, tachyon condensation describes the transition of an unstable D-brane system from an imaginary-mass state to a stable real-mass vacuum [10, 13, 14]. The system starts at a local maximum of the tachyon potential (imaginary mass, $m^2 < 0$) and rolls down to the minimum (real mass, $m^2 > 0$). During this process, the negative energy generated by the tachyon field cancels the positive brane tension, driving the total energy to zero. This restores spacetime supersymmetry. Moreover, spatially inhomogeneous tachyon condensation (e.g., kinks or vortices) can generate lower-dimensional D-branes. This mechanism provides a physical picture for the emergence of our universe from the tachyon sea: a decoherence event triggers condensation, converting imaginary-mass tachyons into real-mass particles and releasing the energy that becomes the hot Big Bang.

4.2 Inverse tachyon condensation from real mass to imaginary mass (from our universe to the tachyon sea)

Inverse tachyon condensation describes the reverse process: real-mass matter, under extreme gravitational conditions, transforms into an imaginary-mass tachyon mode and returns to the vacuum ground state [11, 12]. The mechanism proceeds in four steps: **a) Causal disconnection.** When matter is compressed to Planckian scales, it loses causal contact with the external universe. Its quantum state becomes isolated. **b) Mass sign flip.** Gravitational corrections

invert the sign of the effective mass squared: $m^2 > 0 \rightarrow m^2 < 0$. The system moves from the timelike branch to the spacelike branch. **c) Dimensional increase.** The resulting tachyon modes, through inhomogeneous configurations (kinks/vortices), induce a transition from a lower-dimensional brane to a higher-dimensional D-brane — the inverse of Sen's condensation mechanism. **d) Return to the tachyon sea.** Tachyons re-cohere with the background tachyon sea, converting local classical information into non-local quantum entanglement encoding. Unitarity is preserved. The vacuum energy of the mother sea is defined as zero; the observed dark energy is a residual excitation above this ground state.

4.3 Evolution Mechanism of Imaginary and Real Mass Based on High-Dimensional Spherical Geometry

This section aims to construct a self-consistent dynamical evolution mechanism describing the mutual transformation between imaginary mass (tachyons) and real mass within a high-dimensional spherical (D-brane) model. By introducing the eigenvalues of the spherical Laplacian as boundary constraints, a mass evolution formula isomorphic to the logic of tachyon condensation in string theory is derived, revealing the intrinsic connection between dimensional collapse and the imaginary/real mass transition. In the existing string theory framework, the instability of a D_p -brane originates from the tachyon mode in its open string spectrum, manifesting as an imaginary mass state with $M^2 < 0$. Through deformations of boundary conformal field theory (BCFT), the literature demonstrates that tachyon condensation can lead to a reduction in dimension and, in the new vacuum, eliminate the imaginary mass to produce real mass excitations. This section formalizes this process from a purely geometric-dynamical perspective into an independent high-dimensional spherical evolution model. Consider a sphere S^d of initial dimension d and radius R as the system's foundation [11, 12]. On this sphere, define a tachyon field $T(t, \Omega)$, where Ω denotes the angular coordinates. The system initially sits at a local maximum of the tachyon potential, with an intrinsic mass squared negative: $M_{intrinsic}^2 = -\mu^2$, ($\mu^2 > 0$), corresponding to the imaginary mass phase. To incorporate dimensional effects, expand the field T in spherical harmonics $Y_\ell(\Omega)$:

$$T(t, \Omega) = \sum_{\ell} T_{\ell}(t) Y_{\ell}(\Omega)$$

where ℓ is the angular momentum quantum number. The spherical harmonics satisfy the Laplace equation on S^d :

$$\nabla_{S^d}^2 Y_{\ell}(\Omega) = -\frac{\ell(\ell + d - 1)}{R^2} Y_{\ell}(\Omega)$$

Here $\frac{\ell(\ell + d - 1)}{R^2}$ is the positive definite energy term introduced by spherical geometry, central to the subsequent mass evolution. Within an effective field theory framework, for each mode $T_{\ell}(t)$ we construct the Lagrangian:

$$L = \frac{1}{2} \dot{T}_{\ell}^2 - \frac{1}{2} \frac{\ell(\ell + d - 1)}{R^2} T_{\ell}^2 - V(T_{\ell})$$

where $V(T_{\ell})$ includes the $-\mu^2$ tachyon potential. Varying T_{ℓ} yields the equation of motion:

$$\ddot{T}_{\ell} - \frac{\ell(\ell + d - 1)}{R^2} T_{\ell} + V'(T_{\ell}) = 0$$

Reorganizing into a standard wave equation, we define the effective mass squared M_{eff}^2 via:

$$\ddot{T}_{\ell} = \left[\frac{\ell(\ell + d - 1)}{R^2} - V''(T_{\ell}) \right] T_{\ell}$$

At the initial moment ($T \approx 0$), $V''(0) = \mu^2$, giving the core formula for imaginary/real mass evolution:

$$M_{eff}^2(R, \ell) = -\mu^2 + \frac{\ell(\ell + d - 1)}{R^2}$$

This formula clearly shows how the nature of the mass evolves with geometric parameters. In the imaginary mass regime, when the sphere radius R is small (Planck-scale) or the angular momentum ℓ is low such that $\frac{\ell(\ell + d - 1)}{R^2} < \mu^2$, we have $M_{eff}^2 < 0$ – the system exhibits imaginary mass tachyons, corresponding to an unstable high-dimensional D-brane. At the critical transition point:

$$R = R_c = \sqrt{\frac{\ell(\ell + d - 1)}{\mu^2}}, \quad M_{eff}^2 = 0, \text{ crossing the boundary between imaginary and real mass.}$$

In the real mass regime, when R continues to increase (spatial expansion) or ℓ grows so that the geometric term dominates, $M_{eff}^2 > 0$ – the imaginary mass disappears and the system evolves into stable real-mass particles. This derivation also reveals the dynamical picture of dimensional reduction: the initial high-dimensional sphere S^d constrains the tachyon via boundary conditions, rendering it imaginary mass; as R evolves or tachyon condensation occurs, the system rolls to a new vacuum. In this new stable state, the perturbative excitation spectrum no longer contains tachyons but exhibits real mass, and the effective dimension may collapse to a lower one (e.g., the three dimensions of interest). In conclusion, by incorporating the geometric constraints of a high-dimensional sphere S^d , this section successfully derives dynamical equations for the mutual transformation between imaginary and real mass. The result shows that imaginary mass is not an exotic exception but a natural consequence of specific geometric conditions (small radius R or low mode ℓ). By adjusting the geometric parameter R , the system smoothly undergoes a phase transition from imaginary mass (unstable high dimension) to real mass (stable low dimension), providing a clear and self-consistent mathematical physics framework for understanding tachyon condensation and the origin of dimensions.

4.4 Critical Temperature of the Imaginary-Real Mass Phase Transition in High-Dimensional Spherical Geometry

This section aims to derive, based on a high-dimensional spherical collapse model, the critical temperature of the phase transition between imaginary and real mass during the evolution of the universe from 4 to 3 dimensions, and to demonstrate the equivalence of this temperature to the characteristic temperature of the cosmic inflation epoch. The entire derivation is based on the principle of Helmholtz free energy minimization in finite-temperature quantum field theory. The system state is described by its radius R , and the equilibrium state is determined by the vanishing

first variation of the free energy $F(R, T)$: $\frac{\partial F(R, T)}{\partial R} = 0$. The free energy $F(R, T)$ consists of three

physical contributions. The first term is the geometric stiffness term A/R^2 , originating from the high-dimensional spherical geometry constraint (Section 5.3 of the reference). Its physical meaning follows directly from the eigenvalues of the Laplacian on a high-dimensional sphere. The

eigenvalue $\frac{\ell(\ell + d - 1)}{R^2}$ of the radial equation is obtained via separation of variables. To maintain

dimensional consistency and simplify the expression, this part is combined into the coefficient A . This term represents the cost of spatial curvature: when the radius R is very small, the space is highly curved and the energy rises sharply, acting like an elastic potential energy resisting compression. The second term is the thermal fluctuation term BT^2R^2 , originating from the high-temperature expansion of quantum field theory (Matsubara formalism). Its physical meaning is to introduce the

temperature T . In finite-temperature QFT, the free energy density in the high-temperature limit generally contains a term proportional to T^4 . Considering that the system volume is proportional to R^d (where d is the dimension), after renormalization or variable substitution, its contribution to the free energy typically takes the form BT^2R^2 (especially near $d=4$). This term represents the expanding force of thermal noise: the higher the temperature T , the more intense the thermal motion, and the larger this term, tending to "inflate" the space. The third term is the imaginary mass driving term $-\mu^2R^2$, originating from the instability of the tachyon sea or high-dimensional state (Sections 3.1 and 5.3 of the reference). Its physical meaning corresponds to the intrinsic imaginary mass squared $-\mu^2$ mentioned in the reference, which drives the system away from the origin towards larger radii. This term represents the inherent expansion tendency of high-dimensional space and is the root cause of system instability before dimensional collapse. Summing the three contributions, the total free energy of the system is:

$$F(R, T) = \frac{A}{R^2} + BT^2R^2 - \mu^2R^2$$

Imposing the equilibrium condition:

$$\frac{\partial F}{\partial R} = -\frac{2A}{R^3} + 2(BT^2 - \mu^2)R = 0$$

Solving for the temperature-dependent radius gives:

$$R(T)^2 = \frac{A}{BT^2 - \mu^2}$$

This equation clearly demonstrates the jump mechanism. In the high-temperature stable regime ($T > T_c$), when the temperature is extremely high (e.g., the Planck temperature T_P in the early universe), the thermal fluctuation term BT^2 dominates over the imaginary mass term μ^2 , making the denominator positive. The system then possesses a finite, stable small-radius solution, corresponding to the "tachyon universe" (high-dimensional, imaginary mass, superluminal) described in the reference. At the critical point ($T = T_c$), the critical temperature is defined by $T_c = \mu/\sqrt{B}$. Then $BT_c^2 - \mu^2 = 0$, leading to $R(T) \rightarrow \infty$. This means that at T_c , the original small-radius vacuum state becomes completely unstable. In the low-temperature regime ($T < T_c$), the denominator becomes negative, so the real solution for R disappears. The system can no longer maintain equilibrium by smoothly adjusting R ; instead, it must undergo a first-order phase transition via quantum tunneling, jumping from an unstable "small universe" to a completely new, macroscopic "large universe". This corresponds to the "slow universe" (three-dimensional, real mass, speed-of-light limit) in the reference. Thus, the critical temperature derived from finite-temperature free energy minimization is $T_c = \mu/\sqrt{B}$, and under Planck-scale boundary conditions this coincides with the Planck temperature, precisely matching the epoch of cosmic inflation. After inflation ends, the radius R freezes and a thermodynamic barrier isolates dimensions, preventing further reactions. Thus, while 4D objects may pass through our universe, the frozen geometry keeps them stable and non-interacting with our 3D space.

Reviewing the critical geometric condition in the tachyon sea model, the effective mass squared M_{eff}^2 of the system is determined by the geometric constraint of a 4-dimensional sphere and the intrinsic imaginary mass:

$$M_{\text{eff}}^2(R, l) = -\mu^2 + \frac{l(l + d - 1)}{R^2}$$

At the boundary between imaginary and real mass, the radius reaches its critical value: $R_c = \frac{\sqrt{l(l+3)}}{\mu}$

According to cosmological principles, the energy density ρ is inversely proportional to the fourth power of the curvature radius R . At the critical point, the energy density ρ_{phase} can be expressed as:

$$\rho_{\text{phase}} \propto \frac{1}{R_c^4} = \frac{\mu^4}{l^2(l+3)^2}$$

Adopting the Planck-scale boundary condition you have set, take $\mu \sim M_P$ (the Planck mass) and the angular momentum mode ℓ on the order of Planck scale ($\ell \sim 1$), the above expression simplifies to: $\rho_{\text{phase}} \approx M_P^4$. This indicates that the energy density at the phase transition is approximately the Planck density. In the hot big bang cosmological model, the energy density in the radiation-dominated era is related to the temperature T by $\rho \propto T^4$. Substituting the critical density yields the phase transition critical temperature T_c : $T_c^4 \approx M_P^4$, Hence: $T_c \approx M_P$. That is, the critical temperature is approximately the Planck temperature $T_P \approx 1.4 \times 10^{32} \text{K}$. This derived critical temperature corresponds exactly to the temperature of the cosmic inflation epoch.[15] The physical mechanism is as follows: in inflationary theory, the universe around the Planck time ($t \sim 10^{-43} \text{s}$) is at an extremely high temperature, near the Planck scale, and may have been in a high-dimensional symmetric state (such as the 4-dimensional sphere in the tachyon sea model). When the universe expands and cools to $T_c \approx M_P$, the radius R of the 4-dimensional sphere precisely expands to the critical value R_c . At this moment, the geometric constraint term $\frac{l(l+3)}{R^2}$ balances the imaginary mass term μ^2 , triggering dimensional collapse (from 4 to 3 dimensions) and the imaginary-real mass transition. This derivation shows that the "dimensional reduction" phase transition in the model occurs, on the cosmological timeline, exactly at the middle-to-late stage of inflation. In summary, the phase transition critical temperature $T_c \approx M_P$ derived from geometric dynamics perfectly matches the temperature of the cosmic inflation epoch in standard cosmology, providing strong quantitative support for the hypothesis that "dimensional origin and inflation share a common source."

5. TOE Framework

5.1 Zero-Energy Principle

In the tachyon sea model, the universe is treated as a closed quantum - gravitational system. Unlike standard cosmology where matter and geometry are separate entities, here they are two facets of the same vibrational mode. We employ the principle of stationary action to derive the zero - energy condition. The total action S of the system is the sum of the geometric (Hilbert) term and the matter (Tachyon) term, integrated over the d -dimensional spatial volume V_d :

$$S = \int_{V_d} R \sqrt{-g} d^d x + \int_{V_d} \left(\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) \right) \sqrt{-g} d^d x$$

where κ_d is the d -dimensional gravitational constant, R is the Ricci scalar (representing the "stiffness" of space), and $V(\phi) = -\frac{1}{2} \mu^2 \phi^2$ is the tachyon potential (negative energy).

By varying the action with respect to the metric $g_{\mu\nu}$, we obtain the d -dimensional Einstein equations (the geometric side):

$$G_{\mu\nu} = \kappa_d T_{\mu\nu}^{(m)}$$

Simultaneously, varying with respect to the field ϕ yields the Klein - Gordon equation (the matter side) in curved space:

$$\square\phi - \frac{\partial V}{\partial\phi} = 0 \implies \frac{1}{\sqrt{-g}}\partial_\mu(\sqrt{-g}g^{\mu\nu}\partial_\nu\phi) + \mu^2\phi = 0 \text{ where, } \square\phi = \left(\frac{\partial^2}{\partial t^2} - \nabla^2\right)\phi$$

In a closed universe (finite volume V_d), general relativity dictates that the total Hamiltonian must vanish. This is not an assumption but a result of the diffeomorphism invariance of the action.

We compute the total Hamiltonian density $H=H_G+H_M$. Substituting the solutions from Chapter 4 (Spherical Harmonics Y_l and radial modes R_n) into the Hamiltonian, we find:

$$H = \frac{1}{2} \int_{\phi}^2 + \frac{1}{2} (\nabla\phi)^2 + V(\phi) + \frac{1}{2\kappa_d} (R - 2\Lambda) = 0$$

To connect with the Tachyon intuition (Section 4.1), we decompose the Hamiltonian into radial (longitudinal) and angular (transverse) parts using the metric of a d-sphere S_d :

Radial (Longitudinal) Part: Corresponds to the Tachyon vibration. The kinetic energy and the tachyon potential $-\frac{1}{2}\mu^2\phi^2$ contribute a negative energy density $\langle H_{rad} \rangle \approx -\mu^2$.

Angular (Transverse) Part: Corresponds to the "Centrifugal" vibration. The spherical Laplacian introduces the term $\frac{l(l+d-1)}{R^2}$. This acts as a positive energy barrier (geometric stiffness)

$$\langle H_{ang} \rangle \approx + \frac{l(l+d-1)}{R^2}.$$

For the system to be stable (oscillatory, not exploding), the energy of the Tachyon must be exactly canceled by the energy of the "geometry" (**Appendix A and Appendix B**):

$$\begin{aligned} \langle H_{rad} \rangle + \langle H_{ang} \rangle &= 0 \\ -\mu^2 + \frac{l(l+d-1)}{R^2} &= 0 \end{aligned}$$

The zero - energy equation is thus derived from first principles. It demonstrates that matter (negative tachyon energy) and geometry (positive curvature/centrifugal energy) are locked in a perfect balance. The "universe" is simply the region where these two cancel out, satisfying $E_{total}=0$. More generally, the first principle of the universe under the tachyon sea hypothesis is: **All stable energy must be confined by its own geometric envelope, and the total energy of the two is zero.**

$$\langle \hat{H}_g \rangle + \langle \hat{H}_m \rangle = 0$$

where $\langle \hat{H}_g \rangle$ is the average expectation energy of the geometric (gravitational) part, and $\langle \hat{H}_m \rangle$ is the average expectation energy of the matter part (including fields, particles, dark matter, etc.). The expectation value of the total energy (gravitational energy + matter energy) of the universe is zero. The microscopic equation is the Wheeler-DeWitt equation: $\hat{H}|\psi\rangle = 0$. In this framework, the geometric energy E_{geo} is not a monolithic entity but arises from distinct physical degrees of freedom of the flexible spacetime manifold. We decompose it into longitudinal (radial) and angular (torsional) contributions: $E_{geo}=E_{rad}+E_{ang}$. Radial Energy component corresponds to the compression, stretching, and convexity changes of the spatial boundary. It is directly tied to the global scale R of the universe and couples to the matter energy to satisfy the zero-energy constraint. For a d -dimensional manifold,

it is given by: $E_{rad} = \sigma \frac{C_d}{R^{d-1}}$.where $\sigma=\pm l$ denotes the convexity (black hole horizon vs. cosmic boundary). This term is responsible for dimensional freezing and the classical gravitational

interaction. **Angular Energy** component describes torsional vibrations, shear deformations, and "patterned" excitations (e.g., spherical harmonics $Y_{\ell m}$ with $\ell \geq 1$). Crucially, while these modes carry information (such as spin and quantum phase), their **mean energy averages to zero**: $\langle E_{ang} \rangle = 0$. These fluctuations do not contribute to the net energy balance but act as carriers of quantum information and give rise to the non-commutativity of geometric operators.

5.2 Space, General Theory of Relativity and Geometry-Matter Bound State Theory

Space is flexible, plastic, and self-adaptively curved in response to internal matter-energy. There is no space without matter (a negatively curved surface would directly collapse); nor is there matter without space (a naked singularity would lead to infinite density). The origin of space is neither a pre-existing empty stage nor an abstract, entropic mathematical illusion. Matter and space are a natural physically indivisible (inseparable whole). The macroscopic continuous spacetime we perceive is precisely the emergent condensate formed by the topological phase transition, mutual fusion, and freezing of countless "microscopic geometric envelopes" at low energies. In short, space is the "coat" that matter has to "wear" in order to exist – a continuous illusion emerging at macroscopic scales from a microscopic discrete geometric structure. The space of our universe appears large, flat, and hard to disturb because all the matter in our universe jointly and stably maintains a region far larger than the visible universe. The space of our universe is a nearly uniform sphere; the boundary of the universe has negative curvature and negative energy, while black holes, regions that have lost causal contact with horizons, and the projections of tachyons passing through our universe have positive curvature and positive energy.

For an open space, the curvature of our universe cooperates with the matter distribution in the most "**energy-efficient**" way. This is the principle of least action in general relativity: given a matter distribution, the choice of spacetime metric makes the total action take a minimum value. In other words, the way spacetime curves is precisely the twisting that yields the "lowest energy". It can also be understood as the local space "compromising" to respond to distant space so as to satisfy the condition of total energy zero. The equation reduces to the classical Einstein field equations of general relativity. The total action is $S = S_{Grav} + S_{Matt}$.

The gravitational part (Einstein-Hilbert action): $S_{Grav} = \frac{c^4}{16\pi G} \int R \sqrt{-g} d^4x$ where R is the Ricci scalar (the scalar curvature of spacetime) and $\sqrt{-g} d^4x$ is the spacetime volume element. The matter part: $S_{Matt} = \int L_{Matt} \sqrt{-g} d^4x$, where L_{Matt} is the Lagrangian density of the matter. Applying the principle of least action, we set the variation of the action with respect to the metric $g_{\mu\nu}$ to zero, $\delta S = 0$. This gives $\frac{\delta S_{Grav}}{\delta g^{\mu\nu}} + \frac{\delta S_{Matt}}{\delta g^{\mu\nu}} = 0$. The first term yields $\frac{c^4}{16\pi G} \left(R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right)$, and the second term is defined as $-\frac{1}{2} T_{\mu\nu}$. Thus, [11] $R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$.

Within the framework of the **Geometry-Matter Bound State Theory**, absolute spacetime acts as a "regulator" that adjusts the internal matter-energy distribution. For an open universe, its curvature and matter distribution synergistically act in the most energy-efficient manner. To establish a consistent framework where spacetime geometry and matter fields achieve dynamical equilibrium, we propose a unified system governed by a master principle that enforces a global duality. The foundation of this framework is the Duality Constraint Equation:

$$C[\phi, g] = 0$$

which enforces the global zero-energy condition, representing the topological "compromise" between the universe's radial (geometric) and angular (matter) degrees of freedom. This constraint is not derived; it is the defining axiom of the theory. The entire dynamics are then compactly encoded in the **Meta-Action**:

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} + L_m(\phi) + \lambda \cdot C[\phi, g] \right]$$

Varying this action yields the coupled system. First, the response of spacetime geometry is given by the Modified Einstein Equation:

$$G_{\mu\nu} = 8\pi G \left(T_{\mu\nu}^{(m)} + \lambda \frac{\delta C}{\delta g_{\mu\nu}} \right)$$

demonstrating that curvature is sourced not only by matter but also by the constraint itself, allowing space to flexibly match the matter content. Conversely, the behavior of the matter field is governed by the Unified Matter Equation:

$$(\square - \mu^2)\phi + \frac{\partial L_{int}}{\partial \phi} = -\lambda \frac{\delta C}{\delta \phi}$$

This single equation unifies the treatment of tachyonic ($\mu^2 < 0$) or real mass ($\mu^2 > 0$) and standard fields, where the stability of both is guaranteed by the balancing force from the constraint. Thus, the constraint $C = 0$, the meta-action S , the geometric equation, and the matter equation form a closed, self-consistent quartet that defines the theory. The precise definitions of all quantities, the explicit form of the constraint C , and the detailed derivation and constraint rules of the equations are provided in Appendix C and Appendix D.

5.3 Space, Force and Symmetry

Special Relativity: Under the framework of the tachyon sea cosmic evolution model proposed in this paper, special relativity emerges as a natural corollary of the zero-energy constraint in flat spacetime, applicable exclusively to inertial frames free from acceleration and massive matter disturbances. In inertial frames A and B moving at constant relative velocity, the absence of real-mass matter pulling on spacetime ensures the space remains an undistorted, flat Minkowski geometry, where the positive geometric energy and internal matter negative energy naturally satisfy the total zero-energy equilibrium, and the physical laws in both frames are completely equivalent. The speed of light c is inherently the characteristic speed of the zero-energy state in flat space: photons have zero rest mass and carry no additional matter negative energy, so their propagation does not break the spatial zero-energy balance, making c invariant across all inertial frames regardless of the observer's motion state—this property is not a postulated assumption but an inevitable consequence of the zero-energy constraint. The coordinate transformation between the two inertial frames must satisfy physical law covariance under the zero-energy constraint while preserving the invariance of the speed of light and spatiotemporal geometric structure; the only linear transformation that meets these conditions is the Lorentz transformation. Effects derived from this transformation, such as length contraction and time dilation, are essentially observable geometric manifestations of space adjusting to maintain zero-energy balance from different inertial observation perspectives. Furthermore, gravitational waves, as intrinsic geometric vibrations of space itself in flat spacetime without relying on matter disturbances, propagate at exactly the same

speed as light c , the natural zero-energy benchmark, so both the speed of light and gravitational propagation speed obey the same Lorentz covariance in inertial frames, sharing the identical zero-energy baseline.

Symmetry: Within the context of three-dimensional $SO(3)$ spatial symmetry inherent to our observable universe, all fundamental interactions—including electromagnetism, the weak force, and the strong force—are not independent entities imposed externally, but rather intrinsic geometric responses of spacetime itself. Driven by the underlying principle of minimizing local energy deviation and ultimately maintaining the zero-energy equilibrium of the spatial domain, these forces emerge as the adaptive curvature or "distortion" of the three-dimensional $SO(3)$ manifold when constrained by the presence of matter carrying corresponding gauge symmetries ($SU(3)$, $SU(2)$, $U(1)$). Regardless of their range, be it the short-range weak and strong nuclear forces or long-range electromagnetism, their manifestations are fundamentally referenced against and dictated by the imperative of preserving the lowest possible energy state within our native three-dimensional spatial fabric, effectively representing the projection and geometric accommodation of high-dimensional or internal symmetries onto the $SO(3)$ spatial framework.

Asymptotic Freedom: Within the framework of the three-dimensional $SO(3)$ spatial zero-energy constraint, asymptotic freedom emerges as a natural geometric consequence rather than an ad hoc quantum effect. As established, the strong force is the adaptive curvature/distortion of the three-dimensional $SO(3)$ manifold, triggered by the presence of quarks carrying $SU(3)$ gauge symmetry, with the sole purpose of canceling the matter energy to preserve the total zero-energy equilibrium of the local spatial domain. At short quark separations, the total matter negative energy perturbation is confined to an extremely small spatial volume; the $SO(3)$ space only needs minimal, localized distortion to offset this perturbation and maintain zero-energy balance, resulting in a very weak effective strong coupling—this is exactly the asymptotic freedom regime where coupling approaches zero as distance tends to zero. At large separations, offsetting the distributed negative energy of the two quarks requires the $SO(3)$ space to sustain widespread curvature distortion across a much larger volume; the cumulative "tension" of this geometric adaptation grows with distance, leading to an increasingly strong effective coupling—corresponding to the confinement phenomenon where quarks cannot be separated arbitrarily.

Origin of Spin: Within the framework of the three-dimensional $SO(3)$ spatial zero-energy constraint, spin is not an independent intrinsic property inherent to particles, but rather a geometric degree of freedom inevitably generated by three-dimensional space itself to satisfy its own rotational symmetry and maintain zero-energy equilibrium. The $SO(3)$ rotational symmetry of three-dimensional space requires that the total energy of the system remain strictly zero under any rotation operation. However, the $SO(3)$ group possesses a unique topological nature: certain quantum states require a 720° rotation to return to their original configuration (corresponding to spinor representations, i.e., the $SU(2)$ covering group). Relying solely on scalar states cannot exhaust all solutions of the zero-energy equation. Therefore, spin is essentially a geometric label forcibly assigned to matter by space to adapt to its own symmetry and prevent energy gaps. For matter in our native universe, the carried spin values ($1/2$, 1 , $3/2$, etc.) are precisely the unified geometric attributes allocated by the three-dimensional $SO(3)$ space to fully preserve rotational symmetry and maintain zero-energy balance. It is not something "grown" by the particles themselves, but rather a "symmetry accessory" rigidly imposed by space, solely to ensure that the total energy remains stable at zero regardless of rotation. This logic also directly derives the spin-statistics theorem: due to the

geometric property that half-integer spin fermions change sign upon a 720° rotation, if two fermions occupy identical states, it leads to wave function sign conflicts and breaks zero-energy equilibrium, thus naturally satisfying the Pauli exclusion principle. Integer spin bosons face no such restriction and can aggregate freely. The entire process requires no additional assumptions and is entirely a natural consequence of the three-dimensional $SO(3)$ space's zero-energy requirement.

In summary, the mathematical and physical structure of the 3D $SO(3)$ Zero-Energy Principle (ZEP) framework is constructed upon four core postulates. These principles aim to unify spacetime dynamics with fundamental interactions under a single geometric logic:

- **The Zero-Energy Postulate:** The total Hamiltonian of the system strictly vanishes, $H_{total} = H_{matter} + H_{geometry} = 0$. This serves as the primary constraint for all physical processes.
- **The Rotational Closure Postulate:** The system maintains zero-energy invariance under $SO(3)$ rotations ($[J_i, H]_{total} = 0$). We propose that spin emerges naturally as the geometric degree of freedom required to preserve this symmetry.
- **The Covariance Postulate:** The fundamental equations are generally covariant, ensuring that the zero-energy constraint holds true across all coordinate transformations (i.e., *Lorentz invariance*).
- **The Minimum Geometric Energy Postulate:** In an open spatial domain, the geometry evolves to minimize its positive energy contribution subject to the zero-energy constraint ($\delta \int L_{geo} d^4x = 0$).

6. Physical Philosophy Framework under the Tachyon Sea Hypothesis

6.1 Decoherence Cutoff

If space is intrinsic to matter, then when the decoherence bubble (our universe) expands, the newly created real particles also bring their own space with them. This process has no inherent stopping mechanism (it does not require the consumption of external resources). The cosmic bubble would expand indefinitely, eventually swallowing the entire tachyon sea. This directly contradicts the assumption that the “tachyon sea is an eternal background”. To prevent an endless, eternal decoherence evolution, the cosmic evolution based on the tachyon sea model requires a decoherence cutoff mechanism: a flat, empty space maintains a total energy of zero, but under the perturbation of vacuum energy (high-dimensional tachyons passing through our universe), it cannot remain flat and uncurved. The curved space couples with the high-dimensional tachyon space, and eventually this spacetime region is torn apart. Cosmic causality (causal relations): causality depends on the observer’s position. The isolated islands that have lost causality are of two types: a) Causal islands: the region enclosed by each observer’s own observable universe boundary (the particle horizon). This boundary is relative and depends on the observer’s position. Different observers have different causal islands, which can overlap. Two mutually unconnected causal islands are bidirectionally isolated. b) Spacelike islands: there exists an absolute, observer-independent spacelike separation between two (or more) events, meaning no causal connection is possible between them. This is not a spatially continuous region but a relational state – an isolated pair or set of events. A black hole is a one-way isolated spacelike island. In this model, within a causal island, the eternal decoherence process must be halted by spatial tearing. When the total matter energy approaches the Planck mass, the local space undergoes a “big rip” rather than a “big freeze”. The curling-up process of the torn space is precisely the inverse tachyon condensation process.

6.2 Vacuum Energy

The energy carried by a single tachyon when it passes through the classical universe. Using the Planck frequency as a cutoff, the estimated vacuum energy density is about $\sim 10^{113} \text{ J/m}^3$, which is of the same order as the Planck energy density from inverse tachyon condensation, $\rho_P = \frac{c^7}{hG^2} \approx 4.6 \times 10^{113} \text{ J/m}^3$. If the vacuum really possessed such a Planck-scale energy density, the universe would immediately be gravitationally bent into an extremely tiny bubble and would never expand as smoothly as it does today. In the tachyon sea cosmic model, vacuum energy is not an intrinsic mysterious energy of “empty space”, but rather the geometric disturbance caused by high-dimensional tachyons passing through the bubble wall and acting on our universe. This energy disturbance manifests only as a Planck-scale geometric energy of space. It is the “ready but not yet released” energy of high-energy, high-dimensional tachyons.

6.3 Black Holes

When matter in our universe is compressed to the extreme, it transforms into tachyons with imaginary mass. These tachyons exist in higher dimensions and move faster than light; they are wrapped in self-adaptive spatial bubbles and quickly escape the black hole, merging into the tachyon sea. From the perspective of the combined system of the tachyon sea and our universe, the total energy is conserved. The curvature of space around a black hole – the black hole continuously “picks up” matter, creating a “debt” of space borrowed from our universe, resulting in a lack of space and a positive-energy curvature of our world. From the viewpoint of matter in our universe alone, the matter is lost, but from the combined perspective of space plus matter energy in our universe, the total energy remains unchanged. The information swallowed by the black hole is encoded in the spatial bubbles of the tachyons and dissolves into the tachyon sea. The whole process simply returns information from our universe back to the tachyon sea. **The role of black holes in the “tachyon sea” model: the “phase-transition recycling station” of the universe.** In the tachyon sea model, the nature of a black hole is not simply a gravitational trap, but a critical device that triggers a dimensional phase transition. It connects the “real-matter universe” with the “imaginary-mass tachyon sea” and is a key link in the cosmic matter cycle. **Core mechanism: phase transition from real mass to imaginary mass. Traditional black hole paradox:** In general relativity, matter falling into a black hole’s singularity seems to lose information forever, violating quantum mechanical unitarity. *Solution in the tachyon model:* Deep inside the black hole (close to, but not necessarily reaching, the singularity), when matter is compressed to Planck-scale energy densities, a fundamental topological phase transition occurs: **Space “melts”:** The 3+1 dimensional continuous spacetime structure of our universe disintegrates, returning to the high-dimensional, discrete “tachyon sea” background. **Matter “reduces”:** Real-mass particles (e.g., protons, electrons) that were originally “frozen” in three-dimensional space lose their spatial confinement, their quantum states decohere, and they transform into imaginary-mass excitations in the high-dimensional tachyon sea.

Resolution of the black hole information paradox: The mainstream holographic principle holds that information is encoded on the two-dimensional event horizon of a black hole, leading to the logical difficulty of “how does information climb from the interior to the surface?” The tachyon model offers an explanation: information is not lost. During the “real $\rightarrow$ imaginary” phase transition, the quantum information of the infalling matter is converted from the encoding mode of “particle states” into the encoding mode of the topological structure of the high-dimensional tachyon sea. The information is encoded **at the phase transition boundary**, not on the event horizon. Due to the

replicability of information, the true “information screen” is stored in the dynamical process of the “phase transition interface” between the real universe and the tachyon sea. This interface is physical, not purely geometric.

The cycle of black holes and the universe: the “breathing” of an open system: The tachyon sea model views the universe as an open system embedded in the background of the “tachyon sea”. The tachyon sea, via low-energy phase transitions (such as inflation), “freezes out” space and real matter, creating and nourishing our universe. Black holes are the key **return channels**: *Internal consumption and dilution*: The universe expands, and matter effectively increases in entropy due to decoherence. *Return (recycling)*: Black holes, through high-energy phase transitions, take the used, highly ordered real matter (e.g., stars) and recycle it, degrading it and returning its information and energy in the form of imaginary mass to the tachyon sea background, providing raw materials for possible new creation. Thus, black holes are not endpoints but converters of matter forms. They resolve the information paradox and provide a dynamic, non-equilibrium picture of cosmic evolution, avoiding the ultimate fate of “heat death”.

6.4 Dark Matter

Because tachyons are dimensionally isolated and of Planck size, they do not participate in the electromagnetic interactions of our universe, nor do they engage in the weak interaction (WIMP); they are purely gravitationally interacting high-dimensional entities.[16] However, their geometric projection in our universe possesses sufficiently high mass, making them ideal candidates for perfect dark matter. Their dynamical evolution from the Big Bang to the present is as follows:

6.4.1 Grand inflation:

Higher-dimensional tachyon-driven inflation model: The inflationary period of the universe can be interpreted as the interaction phase between the high-dimensional tachyon field and the three-dimensional brane universe. The driving force of inflation originates from the potential energy of the high-dimensional tachyon field, which is projected into three-dimensional space via brane coupling, converting into an effective vacuum energy density. The trigger of inflation corresponds to the three-dimensional brane universe moving into a high-density region of the tachyon field in the extra dimensions, where strong tachyon condensation occurs. The superluminal expansion, which appears acausal to a three-dimensional observer, is due to the fact that perturbations of the tachyon field establish causal connections in the extra dimensions (higher-dimensional causality). This model naturally solves the initial conditions problem of inflation: the uniform distribution of the high-dimensional tachyon field guarantees the initial uniformity of three-dimensional space, while the coupling between the brane and the tachyon field automatically provides the negative pressure needed to drive inflation. After inflation ends, the topological defects (e.g., domain walls) produced by the decay of the tachyon field become the initial seeds for the formation of large-scale structure in the universe.

6.4.2 Baryon acoustic oscillations:

Stage 1 – Dark matter builds the skeleton first (end of inflation): The high-dimensional tachyon field itself possesses tiny density inhomogeneities ($\delta\rho/\rho \sim 10^{-5}$). These small “balls” (dark matter candidates) do not interact electromagnetically with photons. This means that before the recombination epoch, when ordinary matter (baryons) is still “locked” by photon radiation pressure and cannot clump, the density perturbations of dark matter can already grow freely under gravity and collapse first, forming the “skeleton” (the cosmic web) of the large-scale structure of the universe – a network of filaments and gravitational potential wells.

Stage 2 – Ordinary matter fills the skeleton: When the universe cools to recombination (about 380,000 years), photons decouple and ordinary matter (gas) becomes free. The gravitational potential wells (the skeleton) already built by dark matter begin to exert their influence. Ordinary matter (gas, dust) “falls” into these pre-existing potential wells under gravity. The ordinary matter further collapses and cools at the centres of the potential wells, forming galaxies and stars. Thus, the distribution of galaxies we see today is actually aligned along the dark-matter skeleton.

Late stage – Mutual locking (our universe holds onto the tachyons): In the present universe (low redshift), structure has become highly nonlinear. **Reversal mechanism:** once ordinary matter has formed high-density galaxies and clusters (objects) in the dark-matter potential wells, these regions generate extremely strong local gravitational potentials. **Locking effect:** this deep gravitational potential well produced by the accumulation of ordinary matter further binds and gathers surrounding dark matter. This forms a positive gravitational feedback loop: dark matter pulls ordinary matter → ordinary matter accumulates and deepens the potential well → the well pulls in more dark matter.

6.4.3 Dark matter fluidity in the post-explosion era:

The projection of high-dimensional tachyons into three-dimensional space has dynamics determined by an effective potential:

$$V_{eff} \approx -\frac{GM}{r} + \frac{1}{2}m_{eff}v^2$$

If m_{eff} is large (the effective mass after compactification of the high-dimensional mass), it becomes “anchored” at the minimum of the gravitational potential (near the centres of galaxies). Therefore, in empty regions of our universe, dark matter flashes by at superluminal speeds; where the gravitational potential is strong, it is dragged and forms “shockwave” effects, leading to accumulation. This effective mass m_{eff} is not a free parameter. In the projected 3D view, the tachyon’s geometric envelope appears as a Planck-scale localized object whose gravitational radius equals its stability boundary, by the zero-energy condition. From the total-energy-zero condition and the stability radius $R_c = z_d/\alpha\ell_P$, the Schwarzschild relation $R_c = 2Gm_{eff}/c^2$ gives $m_{eff} = (z_d/2\alpha)M_P$. For $d=4,5,6$ and $\alpha \sim 1-10$, the mass lies in the narrow range **0.2–2.6 M_P** , i.e. $(0.4-5.7) \times 10^{-8}$ kg. This concrete prediction can be tested by direct detection and by the interferometric phase noise searches of Section 8. Crucially, this m_{eff} is not defined by integrating a density over open 3D space; instead, it characterizes a spatially segregated state—a localized patch with its own envelope, detached from the ambient 3D background. Consequently, the only external handle is its envelope scale R_c , linking mass directly to geometry in a way that mirrors the Schwarzschild form but originates from the discrete structure of space itself.

6.4.4 Mysterious high-energy cosmic rays:

These are often referred to in cosmology as the “GeV/TeV gamma-ray excess”. It is commonly thought that dark matter annihilation is responsible. In regions of the universe with the strongest gravitational potentials (e.g., around supermassive black holes, centres of galaxies), the high-dimensional small “balls” (dark matter) are dragged by gravity and their density becomes extremely high. When two such balls come sufficiently close, according to quantum field theory, they can annihilate – i.e., the two particles collide and “disappear”, converting into standard-model particles (such as quarks, leptons) of our universe. These produced high-energy particles (e.g., quarks) are extremely unstable and quickly decay via strong interactions into a cascade of secondary particles, ultimately producing high-energy gamma rays. These signals have two characteristics:

Signal drowning: the gamma rays produced by such annihilations are mixed with gamma rays from astrophysical processes such as black hole accretion disks and pulsars, making them very difficult to distinguish. *Extremely weak interaction*: the annihilation cross-section (probability of encounter) between high-dimensional balls is very small; even in the highest-density regions, annihilation events are rare.

6.5 Gravity and the Graviton

Gravity originates from the “mechanical-like” vibration of a closed space in response to its internal matter-energy. It is a necessity for the “spatial envelope” to maintain unitary evolution of its quantum state. The vibration consists of two parts: a radial vibration that cancels the energy of the internal matter, and an angular vibration that displays all the internal quantum information. The radial vibration of the gravitational surface is the necessary medium for achieving quantum entanglement. Gravity in the classical universe arises from the spontaneous curvature of otherwise flat space in order to achieve a low-energy state for a closed region, thereby maintaining the lowest-energy configuration of that region. The graviton is not a standard particle; the smallest manifestation of gravity is the spatial curvature energy caused by the projection of a high-dimensional tachyon into our universe. Its size is close to the Planck scale.

6.6 The Illusion of Being at the Center of the Universe and Spatial Flatness

The illusion that we are at the centre of the universe arises because the true boundary of the universe lies far beyond the observable range. The “uniform” expansion of the universe stems from: (1) the geometric energy of flat space tends to zero, so tiny energy perturbations can stretch space; (2) high-dimensional tachyons continue to project into our universe under extreme conditions. The flatness of the observable universe: our universe (3D) is far larger than the observable universe; curvature could be observed at the decoherence wavefront (during inflation) only fleetingly. The cosmic microwave background radiation completely isolates our horizon from deeper cosmic structures.

6.7 Dark Energy

In this model, cosmic scale evolution is driven by the geometric energy of the outer boundary. Since the total energy of flat space is zero, any external perturbation may significantly affect the internal spatial scale. Under this framework, the Hubble constant and dark energy (or equivalently, the dark energy density) are **not fixed constants**, but rather manifestations of boundary or geometric conditions within the observable universe[15]. They can be non-constant, non-monotonic with time, and even exhibit no clear regularity, and may thus be regarded as physical quantities that are not strictly tied to time and can evolve arbitrarily. The geometric properties of space are determined by its own topology and shape: flat space carries no intrinsic geometric energy, while nearly flat space requires only minimal energy to be created or annihilated. Consequently, the observed spatial expansion within the observable universe does not originate from the geometric energy of the three-dimensional space itself; the nature of dark energy ultimately depends on the true cosmic boundary structure far beyond the horizon.

6.8 Three-Dimensional Space in the Classical Universe

Why our universe appears three-dimensional can be understood from the relationship between geometric energy and dimension revealed by the radial spherical Bessel functions: the higher the dimension, the higher the geometric energy of space, making it easier to confine high-energy tachyons. In our lower-dimensional universe, particles adopt a “space-sharing” scheme that confines the binding energies of the strong and weak interactions to extremely small scales. As a

three-dimensional sphere, the geometric energy of our universe is sufficient to independently handle long-range electromagnetic interactions, so there is no need to “split” space to maintain balance.

6.9 Particle Identity (Why Particles Are Identical)

From the diverse, high-dimensional background (tachyon sea) to identical particles in our low-dimensional projection (our universe): in the high-dimensional network of the tachyon sea, there is no unified concept of “space”. The vibrational modes, imaginary masses, and phases of each “tachyon” (node) can be independent and different. Between them there are only quantum correlations, no spatial distances. Therefore, in their native state, they are not identical; they are chaotic “quantum information”. When these high-dimensional nodes are “woven” into our 3+1 dimensional spacetime (spatial envelopes) via a topological phase transition, a fundamental change occurs: **Space sharing**: this “shared space” possesses Lorentz symmetry (rotational and translational invariance). **Symmetry enforcement**: to satisfy the symmetry of this macroscopic space, particles must “disguise” themselves as linear excitations of this space. The energies of particles are classified and interact according to symmetry, and eventually they become identical within this “shared space”. The origin of the Pauli exclusion principle: space is a finite resource. In the low-dimensional projection, position eigenstates form a shared basis. If two fermions attempt to occupy exactly the same spatial quantum state, this would correspond to a superposition error in the low-dimensional projection (high-dimensional information cannot be uniquely mapped), and the system forcibly excludes such ambiguity, manifesting as a repulsive force.

6.10 Intrinsic Properties of Particles and the Cosmological Constant

The tachyon inherent isometry groups, corresponding to rotational symmetries $SO(d)$, coexist as templates. A subsequent cosmic phase transition, triggered by decoherence, nucleates an expanding “classical bubble”—our universe—where the speed of light and causality become effective. Crucially, the high-dimensional geometric symmetries of these tachyons are not spontaneously broken in sequence on a pre-existing stage but are “projected” or “sealed” into the fabric of the nascent spacetime itself during its condensation. This geometric projection, operating through mechanisms analogous to flux-stabilized Kaluza-Klein compactification and chiral zero-mode selection, maps the native $SO(4) \times SO(5) \times SO(6)$ symmetries onto the low-energy gauge structure. Specifically, the $SO(6)$ factor (isomorphic to $SU(4)$) yields $SU(3)_C$ for the strong force and a $U(1)_{B-L}$ symmetry. The $SO(4)$ structure (isomorphic to $SU(2)_L \times SU(2)_R$), in concert with the dynamics of the $SO(5)$ template, undergoes a chiral screening where only the $SU(2)_L$ component couples effectively to the emerging 3+1-dimensional spacetime, naturally explaining the observed chirality of the weak force. The $U(1)_{B-L}$ then mixes with a remnant from the $SU(2)_R$ sector to form the weak hypercharge $U(1)_Y$, completing the electroweak group $SU(2)_L \times U(1)_Y$. The final electroweak symmetry breaking at ~ 100 GeV via a tachyonic condensate (the Higgs field) results in the familiar $SU(3)_C \times U(1)_{EM}$. The intrinsic symmetries of particles in the classical universe (such as charge, colour, flavour, etc.) originate, in essence, from the collapse of rotational symmetries in higher dimensions (or the “tachyon-space-envelope duality”) into the shared three-dimensional space. Because these symmetries can no longer “resonate” with the external space, they become internalised as intrinsic degrees of freedom (T-duality). The $U(1)$ symmetry inside a tachyon inverts and “re-materialises” as the continuous electromagnetic interaction in the external space of the classical universe, and this force cancels with the gravity of the shared space. These intervals are set by the first zeros of the spherical Bessel functions ($z_4 \approx 3.83$, $z_6 \approx 5.14$) that govern stable confinement of the tachyon mode. The spinor covering group of $SO(4)$, $Spin(4)$, is isomorphic

to $SU(2) \times SU(2)$. Under $SO(4) \rightarrow SU(2)$: the rotational symmetry of a 4-dimensional hypersphere, in its spinor representation, splits into two $SU(2)$ factors, one of which corresponds to the weak force. Similarly, $SO(6) \rightarrow SU(3)$: the rotational symmetry of a 6-dimensional hypersphere, in its spinor representation, splits into $SU(4)$, and its projection into three-dimensional space yields $SU(3)$.

Core hypothesis: gauge interactions originate entirely from the geometric structure of the high-dimensional tachyon space. Specifically: The weak force ($SU(2)$ symmetry) originates from the “flipped projection” onto three-dimensional space of angular vibrational modes on the surface of a 4-dimensional sphere (S^3). The strong force ($SU(3)$ symmetry) originates from the “flipped projection” onto three-dimensional space of angular vibrational modes on the surface of a 6-dimensional sphere (S^5).

The strength (coupling constant) of each force is determined by the critical radius of its corresponding high-dimensional sphere. The critical radius is the characteristic scale of stable vibrational modes on the high-dimensional sphere, given by the first zero of the zero-angular-momentum radial Bessel function for that dimension.

Obtaining the critical radii (in natural Planck units): Critical radius for the 4-dimensional tachyon sphere (corresponding to the weak force): $R_w=3.831706$. Critical radius for the 6-dimensional tachyon sphere (corresponding to the strong force): $R_s=5.135622$

Establishing a scaling relation between force strength and radius: In Kaluza-Klein type theories, the gauge coupling constant g is typically proportional to some power of the compactification scale. Based on dimensional analysis and geometric intuition (e.g., force may be related to curvature or energy density, and energy density $\rho \sim 1/R^2$, but the coupling constant might be proportional to the square of the characteristic scale), we make the simplest assumption: **the gauge coupling constant g is proportional to the square of the corresponding critical radius**, i.e., $g \propto R^2$. The physical picture behind this assumption is: a larger critical radius corresponds to higher energy of the vibrational mode, and in the low-energy effective theory, the interaction appears stronger.

Calculating the ratio: Under the above assumption, the ratio of the strong coupling constant to the weak coupling constant is

$$\frac{g_s}{g_w} = \left(\frac{R_s}{R_w} \right)^2 = \left(\frac{5.135622}{3.831706} \right)^2 \approx 1.7965$$

Comparison with experimental values: At low energy scales (e.g., the QCD scale and the electroweak scale), the experimentally measured coupling constants are approximately $g_s \approx 1.22$ and $g_w \approx 0.65$, giving a ratio

$$\frac{g_s}{g_w} \approx \frac{1.2}{0.65} \approx 1.8462$$

The relative error between the theoretical prediction and the experimental value is about 2.7%. Possible sources of error include projection effects from the higher dimensions and curvature effects.

6.11 Quantum Entanglement and Realisation of the Holographic Principle

The selection of a stable physical state within a d -dimensional spherical cavity ($d \geq 3$) is not static but manifests as a continuous dynamic interconversion between the $\ell=1, m=\pm 1$ and $\ell=2, m=0$ configurations. In higher-dimensional topologies, subtle geometric perturbations or topological defects in the background metric induce an accidental near-degeneracy (or accidental degeneracy) between these specific modes. This high-dimensional quantum degeneracy forces the system into a two-level subspace where the $\ell=1, m=\pm 1$ and $\ell=2, m=0$ states become energetically

indistinguishable. Consequently, the system cannot settle into a fixed eigenstate; instead, it undergoes perpetual quantum fluctuations, oscillating coherently between the two. While the $\ell=1, m=\pm 1$ subspace is energetically favorable in a classical sense, its insufficient centrifugal barrier ($\propto 1/r^2$) renders it incapable of preventing wavefunction collapse at the origin. Conversely, the $\ell=2, m=0$ configuration possesses the robust centrifugal potential ($\propto 2d$) necessary to expel probability density and establish a stable, "hollow" profile, albeit at a slight kinetic energy cost. The dynamic switching between these states—driven by the aforementioned near-degeneracy—reconciles this contradiction: the system though quantum fluctuation to occupy the $\ell=2, m=0$ quadrupole configuration to gain centrifugal screening against singularities, only to return to the $\ell=1, m=\pm 1$ mode to restore low-energy favorability. By adopting the radial ground state ($n_r=0$), the principal quantum number is uniquely fixed. This dynamic interconversion—a direct consequence of high-dimensional near-degeneracy—ensures the cavity remains a non-singular, dynamically active entity. It represents the most physically admissible stable excitation, perfectly reconciling low-energy considerations with dynamical stability in arbitrary spatial dimensions

In this model, the tachyon—as an excitation of high-dimensional geometry—manifests a dual reality: internally, it appears as high-energy, imaginary-mass string vibrations (the language of particle physics); externally, it manifests as oscillations of a high-curvature, negative-energy geometric envelope (the language of gravity). This duality is driven by the core topological excitation of dynamic degenerate switching within a d -dimensional geometric sphere, specifically $\ell=1, m=\pm 1 \leftrightarrow \ell=2, m=0$. When countless spheres hosting this bistable oscillation interact, the high-dimensional tachyon sea rapidly evolves into a globally entangled state. The extreme redundancy of higher dimensions erases the concept of "distance," converting local collisions into global topological correlations and forming a natural quantum error-correcting code—information is distributed across the network's collective oscillation modes, and any local perturbation is diluted and repaired by the geometric redundancy. This network provides the physical channel for particle entanglement in our universe: fermions, confined by the Pauli exclusion principle and absolutely separated, can only achieve indirect entanglement via extra-dimensional angular information switching within a shared tachyon mediator, relayed by volume-less bosons. Crucially, this explains the superluminal nature of decoherence: the tachyon sea inherently enables superluminal information updating, allowing entangled particle pairs to undergo instantaneous superluminal decoherence under the strict constraint of angular momentum conservation, without violating low-dimensional causal locality. Ultimately, this mechanism substantiates holographic duality: the gravitational theory in the high-dimensional bulk is equivalent to quantum field theory on the lower-dimensional boundary. ER=EPR is realized as the projection of high-dimensional $\ell=1 \leftrightarrow \ell=2$ dynamics onto our spacetime—microscopic ER bridges are the geometric avatars of high-dimensional quantum entanglement, perfectly reconciling microscopic non-locality with macroscopic causality.[4, 12]

6.12 The Destiny of the Universe – An Eternal Cyclic Symphony

The mechanisms by which matter in our universe returns to the tachyon sea are twofold: **Density compression** – e.g., via black holes. **Density dilution** – when a region (such as a cosmic void) expands to the point that its horizon size becomes smaller than the void, the matter density becomes too low to sustain the geometry of that region, causing it to lose causal contact with the outside. The matter is then forced back into an imaginary-mass tachyon state. Cosmic voids and the cosmic web together compose the grand evolutionary narrative of large-scale structure. As the

horizon gradually shrinks within a void, the particles inside eventually lose causal connection with the surrounding space. When the matter density can no longer sustain the geometric energy of that region, space undergoes a flip and compression, driving all matter into an extremely high-density state, ultimately achieving the reversal from real to imaginary mass and returning to the tachyon sea. Meanwhile, regions of the cosmic web continue to contract, causing matter density to increase; once a critical threshold is exceeded, black holes form. Black holes continuously devour surrounding matter, likewise sending it back to the tachyon sea. After our universe fades away, the next universe is born through a decoherence process within the tachyon sea, beginning a new round of evolution, dissolution, and return – an eternal cyclic symphony.

7. Verification Experiment Proposal

7.1 Atom Interferometer Search for Tachyon Projection Noise

Core Principle: In the tachyon-sea model, high-dimensional tachyons stabilized on 4D and 6D spheres project their angular information into 3D space without energy transfer. This projection imprints a non-Gaussian phase modulation on matter, whose sky-angle distribution can only be decomposed into eigenmodes of the 4D and 6D spherical harmonics. Ultra-deep underground labs provide the low-noise environment needed for long-baseline, long-integration phase measurements.

Optimal Platform: Underground Long-Baseline Atom Interferometers. Conventional dark matter detectors are blind to zero-energy, pure-phase signals. Atom interferometers (e.g., MAGIS-100, MIGA, ZAIGA) directly measure matter-wave phase accumulated over long baselines.[17] Their underground location suppresses seismic, thermal and gravity-gradient noise, allowing integration times from months to years – essential for extracting a weak, persistent non-Gaussian phase signal.

Observables and Data Analysis: For each sky direction, a time series of differential phase is recorded. After subtracting known Gaussian noise (seismic, laser frequency, etc.), the residual phase is examined. Because ordinary noise has vanishing higher-order cumulants, any persistent non-zero bispectrum or trispectrum that varies with pointing direction is a candidate signal. The spatial map of residual phase variance or bispectrum amplitude is then decomposed into spherical harmonics – ordinary 3D harmonics as a baseline, but the signal is predicted to be better described by a finite set of projected eigenmodes of the 4D and 6D spheres (dipole, quadrupole, octupole, hexadecapole). Clear coefficients for these higher-dimensional modes would validate the model.

Immediate Low-Cost Pilot: Existing data from MAGIS-100 (even in commissioning) and auxiliary channels of LIGO/Virgo/KAGRA contain long phase time series with known orientation. Re-analysing these data for non-Gaussian, sky-correlated residuals via higher-order cumulants is a cost-effective first step. A detection would be groundbreaking; a null result (once sufficient sensitivity is reached) would set meaningful upper bounds on tachyon-matter coupling.

Distinguishing from Known Physics: Seismic/thermal noise, gravitational waves, laser frequency noise, conventional dark matter candidates (WIMPs, axions) and the CMB do not produce a zero-energy, non-Gaussian, broadband phase correlation with sky-locked angular structure. Hence such a signature would be unique to the tachyon projection mechanism.

Interpretation: A positive detection would provide first experimental evidence for an imaginary-mass tachyon sea and higher-dimensional stability, validate the duality between string-theory excitations and loop-quantum-gravity envelopes, and reveal a topology-locked

quantum correlation – a direct window into quantum gravity without Planck-scale colliders. A null result at target sensitivity would rule out the simplest version of the model.

7.2 Qubit Test of Angular-Momentum Mediated Entanglement

Principles : In this model, the tachyon is a Planck-scale fundamental unit whose internal structure cannot be directly observed. To an external observer, it manifests as an integrated "spatial vibration packet." Its observable physical degrees of freedom are the collective angular momentum modes of the packet, i.e., the configurations of its collective oscillations characterized by the angular quantum number ℓ (e.g., $\ell=1$ for dipole oscillation or $\ell=2$ for quadrupole oscillation). Quantum entanglement is not a direct interaction between particles; rather, it is a resonant correlation formed when two particles (e.g., fermions) share or couple to the *external* angular vibrational mode of the same tachyon packet. The tachyon sea acts as a medium, where different ℓ modes possess distinct resonant frequencies and selection rules. The establishment and characteristics of entanglement are determined by the specific ℓ mode to which the particles couple.

Existing Evidence: All experimentally successful demonstrations of quantum entanglement implicitly satisfy the angular-degree matching conditions required by this model, making them compatible: a) **Conservation of Angular Momentum:** Any entanglement exchange process strictly obeys angular momentum conservation, e.g., via photon polarization (spin angular momentum) or orbital angular momentum transfer. b) **Bosonic Mediation:** Entanglement is always established via a mediating bosonic field (e.g., photon, phonon). This aligns with the picture that "particles must resonate with the angular mode of a tachyon (a bosonic excitation) to entangle." c) **Mode Matching:** The experimentally observed sensitive dependence of entanglement efficiency on parameters like frequency and polarization can be interpreted as the requirement to match the specific angular resonance mode of a tachyon packet. No established experimental facts contradict this basic picture.

Testable Experiments

Based on the above principles, the following three experiments are designed to verify "angular vibrational mode mediated entanglement" on a superconducting qubit platform:

Experiment 1: Verification of Discrete Spectrum for Angular Mode Resonant Frequencies

Prepare two frequency-tunable superconducting qubits placed in close proximity for coupling. With a fixed drive power, scan the frequency detuning (Δ) between the two qubits and precisely measure the entanglement generation rate at each detuning point. The tachyon model predicts that the entanglement rate will exhibit sharp peaks at specific, discrete detuning values (e.g., a peak at $\Delta=0$ corresponding to $\ell=2$ mode resonance, and peaks at $\Delta=\pm\Omega$ corresponding to $\ell=1$ mode resonance, where Ω is a characteristic frequency), forming a discrete "resonance spectrum." In contrast, standard quantum mechanics, based on a model of continuous electromagnetic field coupling, predicts that the entanglement rate varies smoothly with detuning (e.g., following a Lorentzian lineshape), without such discrete spectral lines.

Experiment 2: Verification of Selection Rules for Angular Momentum Modes (Chirality Test)

Prepare two qubits with identical frequencies. Drive them separately using microwave fields with different circular polarizations (left-handed and right-handed) and measure the resulting entanglement rates. The tachyon model predicts that, due to the specific chiral symmetry of modes like $\ell=2$, one circular polarization will be significantly more effective than the other in exciting this mode, leading to an observable asymmetry in the entanglement rates (i.e., chirality selectivity). In standard quantum mechanics, for symmetrically designed qubits and linear coupling, left-handed and right-handed drives should produce identical entanglement effects.

Experiment 3: Verification of Resonant Dependence of Entanglement Efficiency on Detector Geometric Area

Prepare a series of superconducting qubits with different capacitor pad areas (A) but with key electrical parameters (capacitance C , Josephson energy, etc.) meticulously designed to be strictly identical. Pair each qubit of varying area with a reference qubit of fixed parameters. Under identical conditions, measure the entanglement fidelity (F) for each pair. The tachyon model predicts that the relationship between entanglement efficiency (F) and area (A) is non-monotonic, exhibiting a clear peak—i.e., there exists an optimal area A_{opt} that maximizes entanglement efficiency. The physical explanation is that the coupling of the tachyon's angular information (as a spatial vibration packet) to the qubit has a characteristic matching scale. If the area is too small, the capture efficiency is low; if it is too large, internal interactions cancel each other out and introduce decoherence, causing efficiency to first increase and then decrease. Standard quantum mechanics, with electrical parameters compensated to be identical, predicts that entanglement efficiency should be independent of geometric area, with data showing random fluctuations and no systematic trend.

7.3 First-principles derivation of Physical constants

In this theoretical framework, the bifurcation behavior of the system near the Planck scale and its corresponding characteristic energy provide a natural microscopic origin for the emergence of the Planck constant $\hbar$. $\hbar$ is no longer regarded as a purely fundamental postulate but is instead interpreted as the quantized energy unit that necessarily emerges from the spacetime structure maintaining stability at the minimal scale. This explanatory path—from the first principles of geometric dynamics to the rules of quantization—unifies two seemingly independent fundamental categories: the continuity of geometry and the discreteness of quantization, within a single physical picture. Furthermore, by combining this tachyonic solution with T-duality, the magnitude of the cosmological constant can be naturally derived as well. The successful explanation and recovery of these key constants serve as important evidence of the theory's validity and coherence. Solving the tachyon equations exactly and applying T-duality leads to a derivation of the cosmological constant – this can also be regarded as a key success of the theory.

8. Conclusion

In summary, the tachyon-sea decoherence cosmological model proposed in this paper successfully eliminates the Big Bang singularity, unifies relativity and quantum mechanics, and bridges the dual descriptions of string theory and loop quantum gravity within a single physical picture. The model not only fully reproduces the standard evolutionary course after cosmic inflation, but also provides an internally consistent causal mechanism for a long list of long-standing open problems: the origin of spacetime, the black hole information paradox, the nature of gravity, dark matter, vacuum energy, the origin of quantum entanglement, the identity of particles, and more. The entire universe exhibits a cyclic, renormalising dynamic process, with total energy zero, and gravity and quantum field theory naturally coupled in the dual framework of tachyons and geometric envelopes. The work presented here suggests that the “fully entangled tachyon sea” can serve as a concrete, testable starting point toward a unified theory of quantum information and geometry.

Acknowledgments:

The conceptual model detailed here is the foundational work of the authors. Any research that produces a mathematical formalization or discovers new physical implications based on this

framework must provide proper citation to this original work. The authors express an interest in collaborative development and request that any derivative work, at a minimum, includes an appropriate citation or acknowledgement of this article's authors. Although the mathematical derivation presented herein remains qualitative in nature and lacks rigorous analytical precision, this does not compromise the validity of the underlying physical framework. Further precise derivation of these thresholds is currently under investigation.

Reference

1. Ryu, S. and T. Takayanagi, *Holographic derivation of entanglement entropy from AdS/CFT*. Physical Review Letters, 2006. **96**: p. 181602.
2. Dong, X., D. Harlow, and A.C. Wall, *Reconstruction of bulk operators within the entanglement wedge in gauge-gravity duality*. Physical Review Letters, 2016. **117**: p. 021601.
3. Akers, C., et al. *Geometric constraints from subregion duality beyond the classical regime*. 2016. DOI: 1610.08968.
4. Maldacena, J., *The large N limit of superconformal field theories and supergravity*. Advances in Theoretical and Mathematical Physics, 1998. **2**: p. 231.
5. Bell, J.S., *On the Einstein Podolsky Rosen paradox*. Physics Physique Fizika, 1964. **1**: p. 195.
6. Faulkner, T., M. Li, and H. Liu *A quantum null energy condition from entanglement*. 2017. DOI: 1703.09252.
7. Leutheusser, S. and H. Liu, *Subalgebra-subregion duality: emergence of space and time in holography*. Physical Review D, 2025. **111**: p. 066021.
8. Wheeler, J.A., *The "Past" and the "Delayed-Choice" Double-Slit Experiment*, in *Mathematical Foundations of Quantum Theory*. 1978, Academic Press. p. 9-48.
9. Feinberg, G., *Possibility of faster-than-light particles*. Physical Review, 1967. **159**: p. 1089.
10. Sen, A., *Tachyon condensation on the brane antibrane system*. Journal of High Energy Physics, 1998. **1998**(08): p. 012.
11. Rovelli, C., *Quantum Gravity*. 2004: Cambridge University Press.
12. G't Hooft *Dimensional reduction in quantum gravity*. 1993. DOI: gr-qc/9310026.
13. Sen, A., *Tachyon dynamics in open string theory*. International Journal of Modern Physics A, 2005. **20**(24): p. 5513-5656.
14. Sen, A. and B. Zwiebach, *Tachyon condensation in string field theory*. Journal of High Energy Physics, 2000. **2000**(03): p. 002.
15. Riess, A.G., *Observational evidence from supernovae for an accelerating universe*. Astronomical Journal, 1998. **116**: p. 1009.
16. Smolin, L., *A natural extension of the standard model*. Foundations of Physics, 2018. **48**: p. 1265.
17. Kasevich, M. and S. Chu, *Atomic interferometry using stimulated Raman transitions*. Physical Review Letters, 1991. **67**: p. 181.

Appendix A: Tachyonic Geometric Constraints, Zero-Energy Condition, and Dynamics in (d+1)-Dimensional Spacetime

This appendix aims to provide a complete mathematical derivation for the core arguments in Chapters 4–6 of the main text: **geometric structures spontaneously lock the tachyonic field and ensure its stability**. We will systematically prove, in the context of a d-dimensional space (overall spacetime dimension $d+1$), that: 1) Dirichlet boundary conditions lead to the eigenstate quantization of the static tachyonic field and its radius; 2) The total energy of the system is strictly zero; 3) The spatial curvature distribution exhibits a specific configuration; 4) The lowest-energy excitation corresponds to stable geometric deformations (ellipsoidal vibrations); 5) The observational signatures of such vibrations. The derivation uses the same notation as the main text, e.g., ϕ for the tachyonic field, $V(\phi) = -(1/2)\mu^2\phi^2$ for its potential, $g_{\mu\nu}$ for the metric, and R for the scalar curvature of the d-dimensional space.

A.1 Static Solution, Geometric Constraint, and Stability Criterion

Consider a d-dimensional space (topology S^d) with the spatial part of the metric:

$$ds^2 = dr^2 + r^2 d\Omega_{d-1}^2$$

where $d\Omega_{d-1}^2$ is the standard metric on the (d-1)-sphere. We seek a static, spherically symmetric tachyonic field solution: $\phi = \phi(r)$.

In curved space, the static field satisfies the Klein-Gordon equation:

$$\nabla^2 \phi = \frac{\partial V}{\partial \phi} = -\mu^2 \phi$$

where ∇^2 is the Laplace operator in d-dimensional space. For spherical symmetry, it reads:

$$\nabla^2 \phi = \frac{1}{r^{d-1}} \frac{d}{dr} \left(r^{d-1} \frac{d\phi}{dr} \right)$$

Substituting the potential derivative yields a modified Bessel equation:

$$\frac{1}{r^{d-1}} \frac{d}{dr} \left(r^{d-1} \frac{d\phi}{dr} \right) + \mu^2 \phi = 0$$

Its general solution is:

$$\phi(r) = r^{-(d-2)/2} [AJ_{(d-2)/2}(\mu r) + BY_{(d-2)/2}(\mu r)]$$

where J_ν and Y_ν are Bessel functions of the first and second kind, respectively, with $\nu = (d-2)/2$.

To obtain physically meaningful finite solutions, we impose the following boundary conditions:

1. **Internal Regularity:** The field must be finite at the center $r \rightarrow 0$. Since $Y_\nu(x)$ diverges as $x \rightarrow 0$, we set $B=0$.
2. **Dirichlet Boundary Condition (Geometric Constraint):** At a finite radius $r=R$ (representing the "bubble wall" or decoherence interface), the field is forced to zero: $\phi(R)=0$. This condition mimics the encapsulation and confinement of the unstable field by the flexible geometric structure.

Applying $\phi(R)=0$, we obtain the eigenvalue equation:

$$J_{(d-2)/2}(\mu R) = 0.$$

Let $j_{\nu,n}$ be the n -th positive zero of J_ν . The static solution exists only for discrete values of the geometric radius R :

$$R_n = \frac{j_{(d-2)/2,n}}{\mu} \quad .(A.1)$$

Equation (A.1) is the core stability criterion. The intrinsic instability of the tachyonic field (characterized by the imaginary mass μ) is precisely compensated by the finite spatial size R . A

stable static standing wave forms only when the geometric radius R and the mass parameter μ satisfy this quantization condition. This rigorously proves that **the flexible geometric structure of space is a necessary condition for stabilizing the tachyonic field**. The ground-state ($n=1$) field configuration is:

$$\phi(r) = \phi_0 \frac{J_\nu(\mu r)}{J_\nu(\mu R)} \quad (0 \leq r \leq R).$$

A.2 First-Principles Proof of the Zero-Energy Condition

To verify the total energy is zero, we calculate the integrated energy of the static solution within a d -dimensional ball of radius R . We work in a $(d+1)$ -dimensional Minkowski background metric $ds^2 = -dt^2 + dr^2 + r^2 d\Omega_{d-1}^2$. The energy-momentum tensor for the tachyonic field is:

$$T_\nu^\mu = \partial^\mu \phi \partial_\nu \phi - \delta_\nu^\mu \left[\frac{1}{2} (\nabla \phi)^2 + V(\phi) \right]$$

For the static, spherically symmetric solution, the energy density is:

$$\rho(r) = T_0^0 = \frac{1}{2} \left(\frac{d\phi}{dr} \right)^2 - \frac{1}{2} \mu^2 \phi^2$$

The total energy is obtained by integrating over the entire space:

$$E_{total} = \int_0^R \rho(r) dV = \int_0^R \left[\frac{1}{2} \left(\frac{d\phi}{dr} \right)^2 - \frac{1}{2} \mu^2 \phi^2 \right] dV$$

where $dV = r^{d-1} dr d\Omega_{d-1}$ is the volume element.

Key Step: Using integration by parts and the field's equation of motion, the kinetic term can be rewritten:

$$\int_0^R \frac{1}{2} \left(\frac{d\phi}{dr} \right)^2 dV = -\frac{1}{2} \int_0^R \phi (\nabla^2 \phi) dV + \frac{1}{2} \oint_{S^{d-1}} \phi \frac{d\phi}{dr} dS = \frac{1}{2} \int_0^R \mu^2 \phi^2 dV + \frac{1}{2} \oint_{S^{d-1}} \phi \frac{d\phi}{dr} dS$$

The field equation $\nabla^2 \phi = -\mu^2 \phi$ was used. The boundary term is an integral over the $(d-1)$ -sphere S^{d-1} . Imposing the Dirichlet boundary condition $\phi(R)=0$ and considering regularity at the origin $r=0$, this boundary term vanishes exactly. Therefore,

$$E_{total} = \frac{1}{2} \int_0^R \mu^2 \phi^2 dV - \frac{1}{2} \int_0^R \mu^2 \phi^2 dV = 0$$

This proof is independent of the specific dimension d . **It rigorously establishes that, under Dirichlet boundary conditions, the negative potential energy of the tachyonic field and its gradient kinetic energy (corresponding to geometric confinement energy) cancel exactly, resulting in a system with zero total energy.** This provides a first-principles mathematical foundation for the main text's assertion of a "locked balance between matter (negative energy) and geometry (positive energy)."

A.3 Distribution of the Spatial Curvature Scalar

To investigate how space curves in response to the tachyonic field, we examine the interior geometry via the Einstein equations. In $(d+1)$ -dimensional spacetime, the Einstein equations are:

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = \kappa_{d+1} T_{\mu\nu}$$

Taking the trace gives the relation between the Ricci curvature scalar R and the trace of the energy-momentum tensor $T = T_\mu^\mu$:

$$R = -\frac{2}{d-1} \kappa_{d+1} T$$

For the tachyonic field, the trace is:

$$T = -(d-1) \cdot \frac{1}{2} (\nabla \phi)^2 + d \cdot V(\phi) = -(d-1) \cdot \frac{1}{2} (\nabla \phi)^2 - \frac{d}{2} \mu^2 \phi^2$$

For a concrete analysis, consider a three-dimensional space ($d=3$). Here, the static solution is $\phi(r) = A \sin(\mu r)/r$. Substituting yields:

$$T(r) = - \left(\frac{d\phi}{dr} \right)^2 - \frac{3}{2} \mu^2 \phi^2$$

The curvature scalar $R(r) \propto -T(r)$. Analyzing its distribution:

Near the center ($r \rightarrow 0$): $\phi \approx \text{constant}$, $\nabla \phi \approx 0$, so $T < 0$ and thus $R > 0$, indicating **positive curvature**.

Near the boundary ($r \rightarrow R$): $\phi(R) = 0$, but the gradient $\nabla \phi$ is large, making T negative with a large magnitude, and R positive and significant. This indicates **stronger spatial curvature in the boundary region due to the rapid variation of the field**.

This calculation reveals **how the flexible space curves in response to and encapsulates the tachyonic field**: the interior region has positive curvature, with a more pronounced curvature variation at the boundary. The geometric configuration and field distribution are mutually locked, jointly maintaining the system's zero-energy state.

4 Dynamics of Ellipsoidal Deformation

Breaking perfect spherical symmetry, we consider the lowest-energy excitation—the quadrupole deformation (ellipsoidal shape). Expand the field as:

$$\phi(t, r, \theta) = \phi_0(r) + \epsilon T(t) \cdot Y_{l=2}(\theta)$$

where $Y_{l=2}$ is the d -dimensional spherical harmonic for the quadrupole mode ($l=2$), and ϵ is a small parameter. Substituting into the time-dependent Klein-Gordon equation and separating variables yields the equation for the shape vibration mode $T(t)$:

$$\ddot{T} + \left[\frac{l(l+d-2)}{R^2} - \mu^2 \right] T = 0$$

Defining the vibration frequency ω :

$$\omega^2 = \frac{l(l+d-2)}{R^2} - \mu^2 \quad (\text{A.2})$$

Using the stability criterion for the static solution, $\mu R = j_{(d-2)/2, n}$, and noting that for $l \geq 2$, typically $l(l+d-2) > j_{(d-2)/2, 1}^2$, we find $\omega^2 > 0$. **This means the lowest-energy deformation mode is a stable harmonic oscillation $T(t) \sim \cos(\omega t)$, not a collapse.** This corresponds to the "flexible geometric deformation (membrane vibration)" in the main text. During vibration, the kinetic energy of the geometry and the potential energy of the tachyonic field convert into each other, with the zero-energy condition ensuring $E_{\text{total}} = 0$. Equation (A.2) directly links the geometric characteristic scale R , the field's mass parameter μ , and the vibration frequency ω , **again demonstrating the fundamental constraint geometry imposes on the system's dynamics.**

A.5 Time Evolution and Connection to Experimental Observation

Incorporating the time factor $e^{i\omega t}$, the dynamic curvature scalar also varies periodically:

$$R(t, r, \theta) = R_0(r) + \delta R \cos(\omega t + \varphi)$$

This constitutes the "ripple" of the higher-dimensional geometry. For an external observer in (3+1)-dimensional spacetime, this vibration projects as an effective time-varying Newtonian potential perturbation $\delta U(t)$. In experiments like atom interferometry, this perturbation modulates the matter-wave phase:

$$\delta \phi(t) \approx \frac{1}{\hbar} \int m \delta U(t) dt \propto Y_{l,m}(\theta, \varphi) \cos(\omega t + \varphi_0)$$

The vibration frequency ω , determined by the geometric scale R and mass parameter μ , can reach the order of 10^{43} Hz at the Planck scale. Practical instruments average over this high-frequency signal, registering it as phase noise within a specific bandwidth. This signal may possess anisotropic features (dictated by the higher-dimensional spherical harmonics $Y_{l,m}$) and non-Gaussian statistical characteristics, providing potential fingerprints for experimental identification.

Appendix B: Linear Stability Analysis of the Tachyon-Geometric Bound State

This appendix aims to perform a linear perturbation analysis on the static, spherically symmetric, zero-total-energy "Tachyon-Geometric Bound State" solution proposed in the main text. The core objective is to test the behavior of this equilibrium solution under small dynamical perturbations, determining whether it diverges (unstable) or oscillates (stable). The analysis is strictly based on the mathematical framework established in the main text and Appendix A.

B.1 Background Solution and Perturbation Setup

Background Spacetime: Employ a $D=d+1$ dimensional spacetime metric (1 time, d space dimensions), with the background being the static, spherically symmetric solution:

$$ds_0^2 = -f_0(r)dt^2 + h_0(r)dr^2 + r^2 d\Omega_{d-1}^2$$

where $f_0(r)$ and $h_0(r)$ are determined jointly by the Einstein field equations and the Tachyon field equation ($V(\phi) = -\frac{1}{2}\mu^2\phi^2$), corresponding to the static solution $\phi_0(r)$ derived in Appendix A, which satisfies the Dirichlet boundary condition $\phi(R)=0$, and its associated geometry.

Background Field: The Tachyon field background is this static solution: $\phi_0 = \phi_0(r)$.

Introduction of Perturbations: Introduce arbitrary but small, time-dependent perturbations onto the background solution:

$$\begin{aligned} g_{\mu\nu}(t, r, \theta) &= g_{\mu\nu}^0(r) + \epsilon h_{\mu\nu}(t, r, \theta) \\ \phi_{\mu\nu}(t, r, \theta) &= \phi_{\mu\nu}^0(r) + \epsilon \delta\phi(t, r, \theta) \end{aligned}$$

where $\epsilon \ll 1$ is a small parameter, $h_{\mu\nu}$ is the metric perturbation, and $\delta\phi$ is the field perturbation. This analysis retains terms up to first order in $O(\epsilon)$.

B.2 Total Action and Its Second-Order Variation

The total action of the system is the sum of the Einstein-Hilbert action and the matter field action:

$$S_{\text{total}} = S_{EH} + S_{\phi} = \int d^D x \sqrt{-g} \left(\frac{1}{2\kappa_D} R + L_{\phi} \right)$$

where $\kappa_D = 8\pi G_D$ and $L_{\phi} = \frac{1}{2}g^{\mu\nu}\partial_{\mu}\phi\partial_{\nu}\phi - V(\phi)$. Linear stability is determined by the **second-order variation** of the total action about the background solution.

Second-Order Variation of the Gravitational Part: $\delta^{(2)}S_{EH}$ measures the "elastic potential energy" of spacetime geometry deviating from the background. Standard computation yields:

$$\delta^{(2)}S_{EH} = \frac{1}{2\kappa_D} \int d^D x \sqrt{-g_0} h_{\mu\nu} \varepsilon^{\mu\nu\rho\sigma} h_{\rho\sigma}$$

where $\varepsilon^{\mu\nu\rho\sigma}$ is a second-order differential tensor operator constructed from the background metric $g_{\mu\nu}^{(0)}$ and its derivatives.

Second-Order Variation of the Matter Part: $\delta^{(2)}S_{\phi}$ includes the kinetic and potential energies of the field perturbation itself, as well as the interaction energy between the perturbation and the background field:

$$\delta^{(2)}S_{\phi} = \int d^D x \sqrt{-g_0} \left[-\frac{1}{2} \nabla_{\mu} \delta\phi \nabla^{\mu} \delta\phi + \frac{1}{2} \mu^2 (\delta\phi)^2 \right] + \text{coupling terms}$$

The crucial coupling terms are embodied in the first-order variation of the energy-momentum tensor $\delta T_{\mu\nu}$, which depends linearly on $h_{\mu\nu}$ and $\delta\phi$. This is the mathematical manifestation of the **self-regulating feedback mechanism**: the field perturbation $\delta\phi$ alters the local energy-momentum distribution ($\delta T_{\mu\nu}$), thereby driving a response in the spacetime geometry ($h_{\mu\nu}$); conversely, the

geometric deformation ($h_{\mu\nu}$) also reacts back on the field evolution through geodesic deviation effects.

Coupled Linear Perturbation Equations: Varying the total action $\delta^{(2)}S_{total}$ with respect to $h_{\mu\nu}$ and $\delta\phi$ separately yields a set of coupled linear equations:

$$\text{Geometric Perturbation Eqn: } \varepsilon_{\mu\nu\rho\sigma} h^{\rho\sigma} = \kappa_D \delta T_{\mu\nu}(\phi_0, \delta\phi)$$

$$\text{Field Perturbation Eqn: } (\square_0 - \mu^2)\delta\phi = S(h_{\alpha\beta}; \phi_0)$$

Here, $\square_0$ is the d'Alembertian operator on the background metric, and S is the **source term** generated by the metric perturbation $h_{\alpha\beta}$ in the field equation. The two equations must be solved simultaneously.

B.3 Separation of Variables and the Eigenvalue Problem

The static, spherical symmetry of the background allows for a spectral decomposition of the perturbations, transforming the stability problem into an eigenvalue problem.

1. **Harmonic Time Dependence and Angular Decomposition:** Given the static background, we can assume perturbations have a harmonic time dependence $e^{-i\omega t}$. Furthermore, exploiting spherical symmetry, perturbations are expanded in terms of d-dimensional spherical harmonics $Y_{\ell m}(\theta)$. To investigate the most unstable modes, one typically first analyzes the **spherically symmetric "breathing mode"** (i.e., $\ell=0$ scalar-type perturbations):

$$h_{\mu\nu}(t, r) = e^{-i\omega t} \tilde{h}_{\mu\nu}(r)$$

$$\delta\phi(t, r) = e^{-i\omega t} \tilde{\chi}(r)$$

2. **Reduction to a Radial Eigenvalue Problem:** Substituting the above forms into the coupled perturbation equations and using constraints (e.g., the Regge-Wheeler gauge) to eliminate non-dynamical degrees of freedom, the complex tensor equations can be reduced to one or a set of **coupled second-order ordinary differential equations** for the radial functions $\tilde{\chi}(r)$ and certain metric perturbation components (e.g., $\tilde{h}_{tt}(r)$, $\tilde{h}_{rr}(r)$). After suitable variable choices and coordinate transformations (such as introducing the tortoise coordinate $dr_* = \sqrt{h_0/f_0} dr$), the system can be written compactly as:

$$\left(\frac{d^2}{dr_*^2} + \omega^2 \right) \begin{pmatrix} \tilde{\chi} \\ \tilde{h} \end{pmatrix} = V(r; \phi_0, f_0, h_0) \begin{pmatrix} \tilde{\chi} \\ \tilde{h} \end{pmatrix}$$

Here, $V(r)$ is a 2×2 **potential matrix**, all of whose matrix elements are determined by the static background solution $\phi_0(r)$, $f_0(r)$, $h_0(r)$ and their derivatives. This equation is analogous in form to a coupled-channel Schrödinger equation in quantum mechanics, with ω^2 as the eigenvalue.

B.4 Stability Criterion and Physical Conclusion

The dynamical stability of the system is entirely determined by the nature of the eigenvalue problem for ω^2 :

If $\omega^2 > 0$ for all perturbation modes, then ω is real, and perturbations represent **stable oscillations** about the equilibrium position (i.e., eigenmodes are $e^{-i\omega t}$ with real ω).

If there exists any mode with $\omega^2 < 0$, then ω is imaginary, and the perturbation will grow **exponentially** in time ($e^{|\omega|t}$), indicating linear instability of the system.

In the present theoretical context, the following key stabilization mechanisms are at play:

Geometric Confinement and Dirichlet Boundary Condition: The background solution $\phi_0(r)$ derived in Appendix A is itself an eigenstate of the Tachyon field within a finite geometry (radius R). The Dirichlet boundary condition $\phi(R)=0$ provides a powerful external constraint, suppressing unlimited growth of the field.

Zero-Energy Condition and Self-Regulating Feedback: The total energy of the system is strictly zero ($E_{total}=0$). This implies that any field perturbation $\delta\phi$ attempting to increase (or decrease) its negative potential energy will necessarily and instantly excite a geometric response $h_{\mu\nu}$ (i.e., deformation of spacetime) of appropriate magnitude and opposite sign, providing a restoring force via the gravitational interaction $\delta T_{\mu\nu}$ to conserve total energy. Mathematically, this manifests as strong off-diagonal coupling terms in the potential matrix $V(r)$.

Therefore, it is physically expected that the intrinsic instability of the Tachyon field ($\mu^2 < 0$) is overcome by the **restorative force of the flexible space**. Mathematically, this predicts that under the influence of the potential matrix $V(r)$, the eigenvalues ω^2 will be **shifted upward from their original negative value ($-\mu^2$) to become positive real numbers**. Solving the eigenvalue problem (typically requiring numerical methods) would yield the specific spectrum of stable oscillation frequencies.

Based on the core physical picture of "spatial flexibility" and the "zero-energy constraint," the linear perturbation theory framework constructed here indicates that the static "Tachyon-Geometric Bound State" derived in Appendix A is not merely an equilibrium solution to the Einstein and Tachyon field equations, but is also expected, upon introducing perturbations, to transform unstable exponential growth modes into stable geometric vibrational modes via the self-coupling feedback mechanism between geometry and matter. However, this theoretical framework is mathematically self-consistent and strongly suggests dynamical stability.

Appendix C: General Variational Principle, Extended Field Equations, and the Zero-Energy Bound State as a Special Case

This appendix presents a first-principles derivation of the coupled geometry-matter system from a constrained action extremum principle. It generalizes the framework of Appendices A and B, demonstrating that the zero-energy bound state emerges as a special, self-consistent solution within a broader class of theories allowing for non-minimal coupling and open systems. The goal is to provide a foundational theoretical basis from which the specific phenomenology discussed in the main text and previous appendices can be derived.

C.1 The Constrained Action Extremum Principle

The fundamental postulate is that the physical evolution of the coupled system—described by the spacetime metric $g_{\mu\nu}$ and the tachyonic scalar field ϕ —is determined by a **constrained extremization of a total action**.

1. **Total Action:** We consider a general action functional defined over a d -dimensional spacetime manifold M :

$$S_{total}[g_{\mu\nu}, \phi, \lambda] = \int_M d^{d+1}x \sqrt{-g} L_{total} + S_{boundary} + S_{constraint}$$

The total Lagrangian density is decomposed as:

$$L_{total} = L_G + L_m + L_{int}$$

$L_G = \frac{1}{2\kappa} R$ is the gravitational Einstein-Hilbert Lagrangian, where $\kappa = 8\pi G_{d+1}$ and R is the Ricci

scalar. $L_m = \frac{1}{2} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - V(\phi)$ is the matter Lagrangian for the tachyonic field with potential

$V(\phi) = -\frac{1}{2} \mu^2 \phi^2$. For our universality, $V(\phi) = \frac{1}{2} m^2 \phi^2$ for $m^2 > 0$.

L_{int} is the **interaction Lagrangian** encoding direct, non-minimal couplings between ϕ and curvature invariants (e.g., $f(\phi)R$, $h(\phi)R_{\mu\nu}\nabla^\mu\phi\nabla^\nu\phi$). This term is the generalization beyond standard General Relativity. In the simplest case of minimal coupling, $L_{int}=0$.

$S_{boundary}$ is the Gibbons-Hawking-York boundary term and any necessary matter boundary terms required for a well-posed variational principle.

$S_{constraint} = \int_M d^{d+1}x \sqrt{-g} \lambda C[g_{\mu\nu}, \phi]$ implements a physical constraint via the Lagrange multiplier field $\lambda(x)$. The constraint functional $C[g, \phi] = 0$ embodies a key physical condition on the system.

2. **The Variational Principle:** The physical configurations $(g_{\mu\nu}(x), \phi(x))$ are those that satisfy $\delta S_{total} = 0$ for independent variations of the metric $\delta g_{\mu\nu}$, the field $\delta\phi$, and the Lagrange multiplier $\delta\lambda$. Variation with respect to λ simply enforces the constraint $C=0$.

C.2 Derivation of the Generalized Field Equations

Performing the variations yields the coupled field equations.

1. **Generalized Einstein Equations:** Varying with respect to the inverse metric $g^{\mu\nu}$ gives:

$$\frac{1}{\sqrt{-g}} \frac{\delta(\sqrt{-g} L_{total})}{\delta g^{\mu\nu}} + \lambda \frac{\delta C}{\delta g^{\mu\nu}} = 0$$

After standard computation (and assuming L_{int} is a function of ϕ , R , and possibly $R_{\mu\nu}$), this takes the form of a modified Einstein equation:

$$G_{\mu\nu} = \kappa \left(T_{\mu\nu}^{(m)} + T_{\mu\nu}^{(int)} \right) + \kappa \lambda T_{\mu\nu}^{(C)} \quad (C.1)$$

Here, $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$ is the Einstein tensor.

$T_{\mu\nu}^{(m)} = -\frac{2}{\sqrt{-g}} \frac{\delta(\sqrt{-g})L_m}{\delta g^{\mu\nu}} = \nabla_\mu \phi \nabla_\nu \phi - g_{\mu\nu} \left[\frac{1}{2}(\nabla\phi)^2 + V(\phi) \right]$ is the canonical stress-energy tensor of

the ϕ field. $V(\phi) = -\frac{1}{2}\mu^2\phi^2$ or $V(\phi) = \frac{1}{2}m^2\phi^2$.

$T_{\mu\nu}^{(int)} = -\frac{2}{\sqrt{-g}} \frac{\delta(\sqrt{-g})L_{int}}{\delta g^{\mu\nu}}$ is the stress-energy contribution from the non-minimal interaction term.

Its specific form depends on the choice of L_{int} .

$T_{\mu\nu}^{(C)} = -\frac{2}{\sqrt{-g}} \frac{\delta(\sqrt{-g})C}{\delta g^{\mu\nu}}$ is the geometric contribution arising from the constraint.

2. **Generalized Tachyon/ Field Equation:** Varying with respect to ϕ gives:

$$\frac{2}{\sqrt{-g}} \frac{\delta(\sqrt{-g})L_{total}}{\delta\phi} + \lambda \frac{\delta C}{\delta\phi} = 0$$

This yields a modified Klein-Gordon equation:

$$\square\phi - \mu^2\phi + \frac{\partial L_{int}}{\partial\phi} - \nabla_\mu \left(\frac{\partial L_{int}}{\partial(\nabla_\mu\phi)} \right) + \lambda \frac{\delta C}{\delta\phi} = 0 \quad (C.2a)$$

$$\square\phi + m^2\phi + \frac{\partial L_{int}}{\partial\phi} - \nabla_\mu \left(\frac{\partial L_{int}}{\partial(\nabla_\mu\phi)} \right) + \lambda \frac{\delta C}{\delta\phi} = 0 \quad (C.2b)$$

The terms involving L_{int} and C represent explicit forces on the field due to geometry and the system constraint, beyond the minimal coupling contained in the covariant d'Alembertian $\square$.

C.3 The Zero-Energy Bound State as a Critical Solution

The specific physical scenario of a self-contained, stable bound state is recovered by imposing a particular **energy closure constraint**.

1. **The Global Energy Constraint:** For a static, spherically symmetric configuration, we demand the system's total gravitating mass-energy (e.g., the ADM mass or the quasi-local mass within a radius R) vanishes. This can be encoded in a constraint functional. A convenient choice for a ball of coordinate radius R is:

$$C[g, \phi] = \int_0^R dr \int d\Omega_{d-1} (NH + N^i H_i) - 0$$

where H and H_i are the Hamiltonian and momentum constraints of General Relativity, and N, N^i are the lapse and shift. In the static gauge and for spherical symmetry, this essentially imposes that the integral of the energy density over the ball, computed from the full stress-energy tensor (including gravitational binding energy), is zero. For the minimal coupling case ($L_{int}=0$), this reduces precisely to the condition $E_{total}=0$ used in Appendix A.2.

2. **Recovery of the Appendix A Solution:** Impose the constraint $C=0$ via the Lagrange multiplier λ . Assume minimal coupling ($L_{int}=0$) and a static, spherically symmetric ansatz for both metric and field. The field equation (C.2) reduces to the Helmholtz-type equation $\nabla^2\phi = -\mu^2\phi$ within the ball. The Einstein equations (C.1), with the energy constraint, determine the background metric. In the weak-field limit (or for a test field in a fixed Minkowski background), the metric becomes approximately flat, and the system reduces to the one solved in Appendix A. The boundary condition $\phi(R)=0$ emerges not as an ad hoc requirement but as a **consequence of the field equation and the energy constraint** applied at a finite boundary. The static, regular solution is again:

$$\phi_0(r) = \phi_0 \frac{J_{(d-2)/2}(\mu r)}{J_{(d-2)/2}(\mu R)}, \text{ with } J_{(d-2)/2}(\mu R) = 0$$

$$\text{Or } \phi_0(r) = \phi_0 \frac{J_{(d-2)/2}(mr)}{J_{(d-2)/2}(mR)}, \text{ with } J_{(d-2)/2}(mR) = 0$$

The Lagrange multiplier λ adjusts self-consistently to satisfy the zero-energy condition globally. This demonstrates that the zero-energy bound state is a **critical solution** of the constrained action principle.

3. **The "Spring Constant" and Geometric Feedback:** The constraint $C=0$ plays the role of a global constitutive relation. The Lagrange multiplier λ can be interpreted as a measure of the tension or pressure required on the boundary to maintain the zero-energy condition. In the equation for perturbations (Appendix B), this contributes to the effective potential matrix $V(r)$, providing the **restoring force** that stabilizes oscillations. The condition $J_{(d-2)/2}(\mu R)=0$ thus appears as a **quantization condition for stable equilibrium**, linking the field's internal scale (μ^{-1}) to the geometric scale (R).

C.4 Extension to Open Systems and the Energy Spectrum

The framework naturally generalizes to non-zero energy states, describing an "open" or "excited" system.

1. **Relaxing the Constraint:** Instead of $C=0$, we can impose $C=E_0$, where E_0 is a fixed (non-zero) energy. This represents a system in contact with a reservoir that fixes its total energy. The variational principle now seeks an extremum of the action subject to this fixed energy.

2. **The Generalized Static Solution:** Solving equations (C.1) and (C.2) with the constraint $C=E_0$ will yield a one-parameter family of static, spherical solutions parameterized by E_0 . The field profile $\phi(r;E_0)$ and the concomitant metric will differ from the zero-energy case. The boundary condition at $r=R$ will generally be more complex than a simple Dirichlet condition, potentially involving a non-zero field derivative, reflecting a balance between internal field stress and boundary pressure/tension.

3. **Energy Spectrum and Stability:** As E_0 varies, we obtain a **spectrum of equilibrium configurations**. The zero-energy solution ($E_0=0$) is expected to be the ground state of this spectrum. Solutions with $E_0>0$ (positive total energy) or $E_0<0$ (negative total energy relative to infinity) are excited states. The linear stability analysis of Appendix B, when applied to these generalized solutions using the equations derived from (C.1) and (C.2), will determine their stability. The zero-energy ground state is anticipated to be stable, while excited states may possess unstable modes, corresponding to decay to the ground state via radiation or other processes.

4. **Coupling Constant and Observables:** The interaction Lagrangian L_{int} , if present, introduces new dimensionless coupling constants (e.g., $\xi_{in} \xi \phi^2 R$). These constants modify the relation between μ , R , and the energy E_0 . They also affect the predicted vibration frequencies ω (from perturbation analysis) and the detailed anisotropic signatures in experimental probes. Thus, the general theory is falsifiable and predictive.

C.5 Radial Reduction under Static Spherical Symmetry

To obtain explicit static, spherically symmetric bound-state solutions, we reduce the general field equations (C.1) and (C.2) under the following physically motivated assumptions:

1. **Physical Setup:**

Consider a real massive scalar field with potential $V(\phi)=+\frac{1}{2}m^2\phi^2$.

Assume minimal coupling, i.e., $L_{int}=0$.

The global zero-energy constraint is enforced via boundary conditions.

2. Symmetry Ansatz:

The metric is static and spherically symmetric:

$$ds^2 = -A(r)dt^2 + B(r)dr^2 + r^2 d\Omega_{d-1}^2$$

where $d\Omega_{d-1}^2$ is the standard metric on the $(d-1)$ -sphere.

The scalar field depends only on the radial coordinate: $\phi=\phi(r)$.

3. Reduced Field Equations:

Scalar Field Equation: Equation (C.2) reduces to

$$\frac{1}{\sqrt{AB}r^{d-1}} \frac{d}{dr} \left(\sqrt{\frac{A}{B}} r^{d-1} \frac{d\phi}{dr} \right) + m^2 \phi = 0 \quad (C.3)$$

Einstein Equations: It is convenient to introduce a "mass function" $M(r)$ defined by

$$B(r) = \left[1 - \frac{2\kappa_d M(r)}{(d-2)r^{d-3}} \right]^{-1}$$

where $\kappa_d=8\pi G_{d+1}$. The tt and rr components of the Einstein equations (C.1) then yield, respectively:

$$\frac{dM}{dr} = \frac{\Omega_{d-1}}{2} r^{d-2} \left[\frac{1}{B} \left(\frac{d\phi}{dr} \right)^2 + m^2 \phi^2 \right] \quad (C.4)$$

$$\frac{A'}{A} = \frac{2\kappa_d M(r) + \kappa_d r^{d-1} \left[\frac{1}{B} \left(\frac{d\phi}{dr} \right)^2 - m^2 \phi^2 \right]}{(d-2)r^{d-3} \left[1 - \frac{2\kappa_d M(r)}{(d-2)r^{d-3}} \right]} \quad (C.5)$$

where Ω_{d-1} is the area of the unit $(d-1)$ -sphere and a prime denotes a derivative with respect to r .

4. Boundary Conditions:

Regularity at the Origin: As $r \rightarrow 0$, we require $\phi(0)$ to be finite, $\phi'(0)=0$, $M(0)=0$, and $A'(0)=0$.

Boundary of the Bound State: At a finite radius $r=R$, we impose the zero-energy condition and the vanishing of the field:

$$M(R)=0, \phi(R)=0. \quad (C.10)$$

Equations (C.4)–(C.10) form a set of coupled ordinary differential equations for the functions $\phi(r)$, $M(r)$, and $A(r)$. They constitute the foundation for numerically solving for the static, spherically symmetric background solution of the bound state.

C.6 Conclusion: A Unified Framework

This appendix has elevated the specific model of Appendices A and B into a comprehensive theoretical framework based on a constrained action principle. The key results are:

1. The coupled geometry-matter dynamics, including potential beyond-GR interactions, derive from the extremization of a general action subject to a physical constraint (e.g., on total energy).
2. The remarkable **zero-energy tachyonic bound state**—with its Dirichlet boundary condition, quantized radius, and proven stability—emerges as the unique ground state solution when the constraint imposes zero total mass-energy.
3. The framework naturally extends to **open systems** with non-zero energy, yielding a spectrum of equilibrium configurations whose stability can be systematically analyzed.
4. The **geometric feedback** mechanism, mathematically expressed through the constraint and the coupling in the field equations, is identified as the fundamental origin of stability, turning the tachyonic instability into a spectrum of stable vibrational modes.

Therefore, the specific results presented in the main text and in Appendices A and B are not isolated phenomena but are the foundational predictions of a broader, self-consistent theoretical structure. This structure provides a principled basis for further exploration, including the detailed study of excited states, the inclusion of additional fields, and the computation of precise observational signatures for experimental tests.

Appendix D: Spacetime-Matter Topological Duality and the Interaction Dynamical Mechanism

D.1 Introduction: The Unity of Spacetime and Matter

Classical physics often decomposes physical systems into two independent parts: the "background spacetime" and the "dynamic matter." However, from a deeper perspective, spacetime geometry and matter fields constitute an inseparable dual system. The theoretical framework presented here indicates that any physical system must simultaneously satisfy two distinct types of constraints: continuous geometric deformation (**longitudinal**) and discrete topological structure (**angular**). These two sets of degrees of freedom are not independent but are interwoven, collectively determining the evolution of the universe.

D.2 Decomposition of Degrees of Freedom: Longitudinal Flexibility and Angular Rigidity

To achieve self-consistent coexistence, the total action of a physical system must incorporate the dual attributes of geometry and matter. We decompose the system's degrees of freedom into two orthogonal dimensions:

Longitudinal Dimension (Longitudinal/Flexible): Continuous Geometric Response

Physical Image: Imagine spacetime as an elastic membrane, its scale described by a radius R or the metric $g_{\mu\nu}$.

Mathematical Property: Continuity. The expansion/contraction of space is smooth, able to change continuously from R_0 to $R_0 + dR$. This corresponds to the spacetime curvature described by General Relativity.

Dynamical Role: When the energy density of internal matter changes, space responds through continuous expansion or contraction. This regulatory mechanism is very "cheap" as it does not involve changes in topological structure. On cosmological scales, this means space can adapt to minute energy fluctuations by changing its radius R , manifesting as the expansion or contraction of the universe.

Angular/Toroidal Dimension (Angular/Rigid): Discrete Topological Constraints

Physical Image: Matter moves on this elastic membrane, or space itself possesses a winding structure (e.g., cosmic strings, spin networks).

Mathematical Property: Discreteness. Angular motion is strictly protected by topology. One cannot have "half a winding" or "1.5 loops." This directly leads to the emergence of quantization conditions—angular momentum $L = n\hbar$, magnetic quantum numbers m must be integers.

Dynamical Role: Angular degrees of freedom cannot be adapted via fine-tuning. To change a system's angular quantum number, a discontinuous quantum jump (energy absorption/emission) must occur. On cosmological scales, this means imparting overall rotation to the entire universe requires overcoming enormous centrifugal forces and altering its topological structure, which is energetically extremely costly.

D.3 Symmetry Breaking and Cosmological Observations

Applying the above principles to our observed universe leads to a striking conclusion: on cosmological scales, longitudinal degrees of freedom are "cheap" and active, while angular degrees of freedom are "expensive" and frozen.

Activity of the Longitudinal: Space is extremely flexible longitudinally. Therefore, when the total energy of the universe experiences minor fluctuations, the system's preferred regulatory method is to change the radius R . This is the observed expansion of the universe—a low-energy, continuous mode of regulation.

Freezing of the Angular: To induce overall rotation in a universe with a radius $R \sim 10^{26}m$ would require energy far exceeding the total energy of the universe. Furthermore, rotation would generate intense centrifugal forces, disrupting the isotropy of the Cosmic Microwave Background (CMB) radiation. Consequently, under the principle of minimum energy, angular degrees of freedom are "frozen" in the lowest energy state, $n=0$ (no rotation). The absence of observed macroscopic rotation is not due to a lack of angular momentum, but because the cost of activating angular degrees of freedom is prohibitive.

Within the unified framework, fundamental interactions differ in how their range couples to the global geometry of the universe. The photon, as a massless gauge boson, mediates the **long-range** electromagnetic force. Its propagation and energy (frequency) are directly modulated by the dynamics of the cosmic scale factor $R(t)$, as manifested in cosmological redshift; thus, it is **strongly coupled** to the longitudinal stretching of the background spacetime. In contrast, the weak force mediated by the $W^\pm$ and Z bosons—which acquire mass via the Higgs mechanism—and the strong force (mediated by gluons), which is short-ranged due to color confinement, are all **short-range forces**. They possess an intrinsic mass scale or confinement energy scale, restricting their interactions to microscopic, local domains. Consequently, they are largely **decoupled** from the cosmological evolution of $R(t)$. This distinction fundamentally corresponds to the continuous, global **longitudinal degrees of freedom** versus the discrete, local **angular degrees of freedom**.

D.4 Decoupling Mechanism of Interaction

The coupling between spacetime and matter is not mediated by a specific "interaction force" but is realized through a global zero-energy constraint. Mathematically, this constraint manifests as a topological boundary condition, requiring the system to balance longitudinal deformation and angular excitation.

Weak Coupling Limit (Everyday and Cosmological Scales):

When the energy density of the matter field ϕ is low, or spacetime curvature varies gently, the interaction between longitudinal and angular components is weak. The system can then decouple into two independent equations:

The geometric part reduces to the Einstein Field Equations, describing how spacetime minimizes energy via continuous scaling.

The matter part reduces to the Klein-Gordon or Dirac equations, describing quantum behavior of particles on a fixed background.

Strong Coupling Limit (Black Holes, the Early Universe, and the Tachyon Sea Model):

In extreme environments with extremely high matter density or curvature (e.g., black hole interiors, the early universe), the boundary between longitudinal and angular degrees of freedom dissolves, and the system enters a highly correlated, strong-coupling state. Here, the classical descriptions of matter and geometry fail; the spacetime background undergoes violent fluctuations, and the topological nature of the matter field becomes fundamentally intertwined with the flexible deformations of geometry.

In this regime, the equations describing the system's evolution reduce to the **Tachyon Sea Model** proposed in this paper. In this model, the dominant matter field consists of the **Tachyon Field** with imaginary mass. The core dynamics of the model are described by a set of non-linear evolution equations coupling the imaginary-mass tachyon field ϕ and the dimensional radius R of space (or the "shell"). In this model, the co-evolution of the imaginary-mass tachyon field and the spatial

dimensional radius jointly depicts a non-singular physical picture of the high-energy state of the early universe or the core region of a black hole.

D.5 Conclusion

The physical reality we observe is a dynamic balance between spacetime geometry (longitudinal/continuous) and matter fields (angular/discrete) under a global zero-energy constraint. The evolution of the universe is the continuous geometric deformation process of this dual system under the premise of conserving topology. Under conventional energy scales, this balance manifests as the separation of gravity and quantum theory. In extreme conditions such as black holes and the origin of the universe, the system evolves into a new physical picture dominated by strong-coupling mechanisms, described jointly by the tachyon field and dynamic geometric radius.